

Tetrahedral sums of Hermitian matrices

Thomas C. Fraser¹, Matthias Christandl¹, and Anton Alekseev²

¹Department of Mathematical Sciences, University of Copenhagen, Universitetsparken 5, 2100 Denmark

²Department of Mathematics, University of Geneva, 7-9 rue du Conseil Général, 1211 Geneva 4, Switzerland

September 4, 2024

Abstract

TODO: write an abstract

1 Introduction

This is the main document for this project. Previous drafts are located in the folder titled “old” and will not necessarily compile properly.

2 Tetrahedra & Hermitian matrices

2.1 Definition

`prob:tetrahedral_problem`

Problem 1 (Tetrahedral eigenvalues). A sextuple $(\alpha, \beta, \gamma, \delta, \epsilon, \phi) \in (\mathbb{R}_{\text{dec}}^d)^{\times 6}$ of candidate eigenvalues are said to be *tetrahedral* if there exists a corresponding sextuple (A, B, C, D, E, F) of $d \times d$ Hermitian matrices satisfying (i) the four linear constraints

$$A + B = C, \quad B + D = F, \quad D + C = E, \quad A + F = E, \quad (1)$$

`{eq:tetrahedral_linear_constraints}`

and (ii) the six eigenvalue constraints

$$\lambda(A) = \alpha, \quad \lambda(B) = \beta, \quad \lambda(C) = \gamma, \quad \lambda(D) = \delta, \quad \lambda(E) = \epsilon, \quad \lambda(F) = \phi. \quad (2)$$

`{eq:tetrahedral_eigenvalue_constraints}`

2.2 Symmetries

Lemma 1 (Shifting). Let $(\tilde{\alpha}, \tilde{\beta}, \tilde{\gamma}, \tilde{\delta}, \tilde{\epsilon}, \tilde{\phi})$ be a given sextuple of candidate eigenvalues and let $x, y, z \in \mathbb{R}$ be given scalars and define shifted candidate eigenvalues by the formula

$$\begin{aligned} \alpha &:= \tilde{\alpha} + x, & \beta &:= \tilde{\beta} + y, & \gamma &:= \tilde{\gamma} + x + y, \\ \delta &:= \tilde{\delta} + z, & \epsilon &:= \tilde{\epsilon} + x + y + z, & \phi &:= \tilde{\phi} + y + z. \end{aligned} \quad (3)$$

`{eq:shifted_tetrahedral_eigenvalues}`

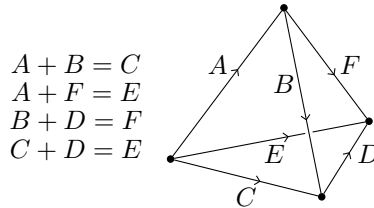


Figure 1: The six Hermitian operators appearing in the tetrahedral eigenvalue constraint problem (Problem 1), and denoted by (A, B, C, D, E, F) above, are subject to four linear constraints corresponding to the four faces of the tetrahedron with coefficients determined by the edge orientations depicted here.

`fig:tetrahedron_labelling_scheme`

Then the shifted eigenvalues $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ are tetrahedral if and only if the unshifted eigenvalues are tetrahedral. Moreover if the unshifted eigenvalues are tetrahedral and (x, y, z) are sufficiently large, i.e.

$$x \geq \min(\tilde{\alpha}), \quad y \geq \min(\tilde{\beta}), \quad z \geq \min(\tilde{\delta}), \quad (4)$$

then the shifted spectra are both tetrahedral and non-negative.

2.3 The $d = 2$ case

In the context of the $d = 2$ tetrahedral eigenvalue problem, each traceless 2×2 matrix, X , may be identified with a three-dimensional vector $(x_1, x_2, x_3) \in \mathbb{R}^3$ using the standard Pauli-matrix basis:

$$X = x_1 \sigma_1 + x_2 \sigma_2 + x_3 \sigma_3 = \begin{pmatrix} x_3 & x_1 - ix_2 \\ x_1 + ix_2 & -x_3 \end{pmatrix}. \quad (5)$$

The eigenvalues of X are then determined by the Euclidean length of the vector (x_1, x_2, x_3) :

$$\lambda(X) = \lambda \begin{pmatrix} x_3 & x_1 - ix_2 \\ x_1 + ix_2 & -x_3 \end{pmatrix} = (\|x\|, -\|x\|), \quad \text{where,} \quad \|x\| = \sqrt{x_1^2 + x_2^2 + x_3^2}. \quad (6)$$

Now, for the special case of the tetrahedral eigenvalue constraint problem, one can assume the sextuple of eigenvalues are determined by a sextuple of non-negative real numbers $(a, b, c, d, e, f) \in \mathbb{R}_{\geq 0}^6$ such that

$$\alpha = (a, -a), \quad \beta = (b, -b), \quad \gamma = (c, -c), \quad \delta = (d, -d), \quad \epsilon = (e, -e), \quad \phi = (f, -f). \quad (7)$$

Using this parameterization, it can be shown that the constraints imposed in Problem 1 precisely describe the space of all possible edge-lengths (a, b, c, d, e, f) of a genuine Euclidean tetrahedron in $\mathbb{R}^3$ [Ric02]; namely, the conjunction of a quadruplet of triangle-inequalities,

$$|a - b| \leq c \leq a + b, \quad |b - d| \leq f \leq b + d, \quad |d - c| \leq e \leq d + c, \quad |a - f| \leq e \leq a + f, \quad (8)$$

together with a single non-linear constraint of the form

$$D(a, b, c, d, e, f) = \frac{1}{288} \det \begin{pmatrix} 0 & 1 & 1 & 1 & 1 \\ 1 & 0 & a^2 & e^2 & c^2 \\ 1 & a^2 & 0 & f^2 & b^2 \\ 1 & e^2 & f^2 & 0 & d^2 \\ 1 & c^2 & b^2 & d^2 & 0 \end{pmatrix} \geq 0. \quad (9)$$

The quantity $D(a, b, c, d, e, f)$ is known as a Cayley-Menger determinant and, when positive, is equal to square of the volume of the corresponding tetrahedron [SJS18]. Figure 2 depicts a three-dimensional projection of the solution space described by Eq. (8) and Eq. (9).

Remark 1. For 2×2 Hermitian matrices it is possible to visualize the set of eigenvalues (in the same (a, b, c, d, e, f) coordinates used above) which are known to be non-tetrahedral on the basis that they violate Eq. (34). Specifically, consider the *minimal* shifting of the traceless eigenvalues in Eq. (7), i.e.,

$$\begin{aligned} \alpha &= (2a, 0), & \beta &= (2b, 0), & \gamma &= (a + b + c, a + b - c), \\ \delta &= (2d, 0), & \epsilon &= (a + b + d + e, a + b + d - e), & \phi &= (b + d + f, b + d - f). \end{aligned} \quad (10)$$

With respect to these coordinates, Eq. (34) becomes

$$\begin{aligned} F(a, b, c, d, e, f) &:= -(a + b + c) \log(a + b + c) - (a + b - c) \log(a + b - c) \\ &\quad - (b + d + f) \log(b + d + f) - (b + d - f) \log(b + d - f) \\ &\quad + (a + b + d + e) \log(a + b + d + e) + (a + b + d - e) \log(a + b + d - e) \\ &\quad + (2b) \log(2b) \geq 0 \end{aligned} \quad (11)$$

See Fig. 3 for a depiction of the region violating Eq. (11).

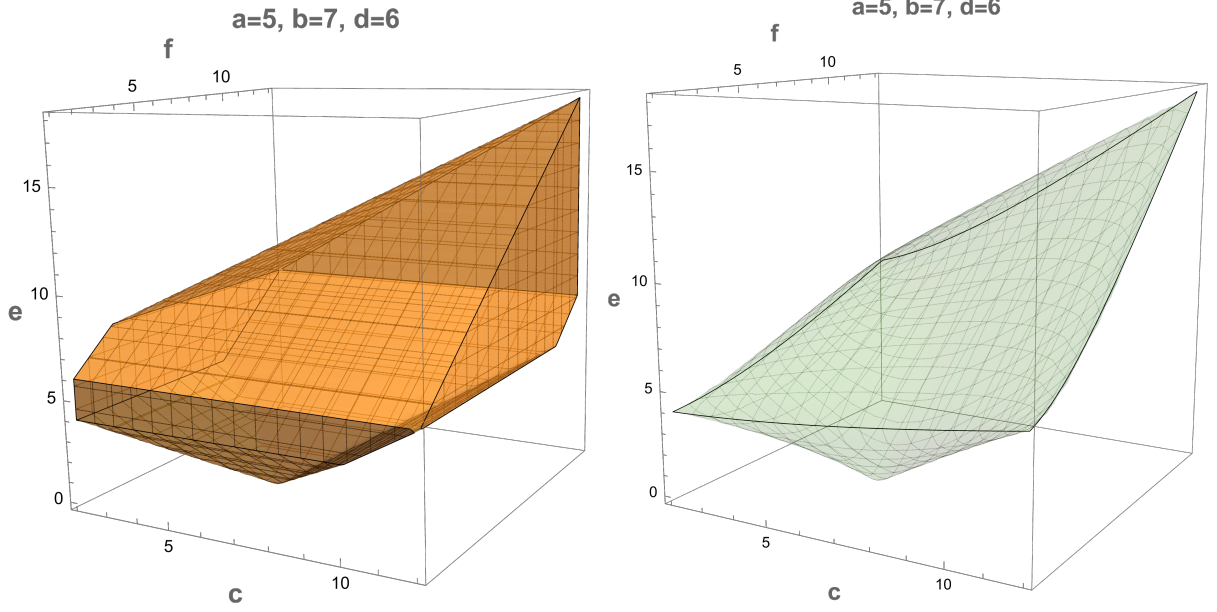


Figure 2: Projections of admissible side-lengths (a, b, c, d, e, f) for Euclidean tetrahedron along the coordinates $(a, b, d) = (5, 7, 6)$ (Left) The polyhedron of values (c, e, f) satisfying the triangle inequalities for each face Eq. (8). (Right) The set of all values (c, e, f) satisfying both Eq. (8) and Eq. (9). [fig:tetrahedron_moduli_space_example](#)

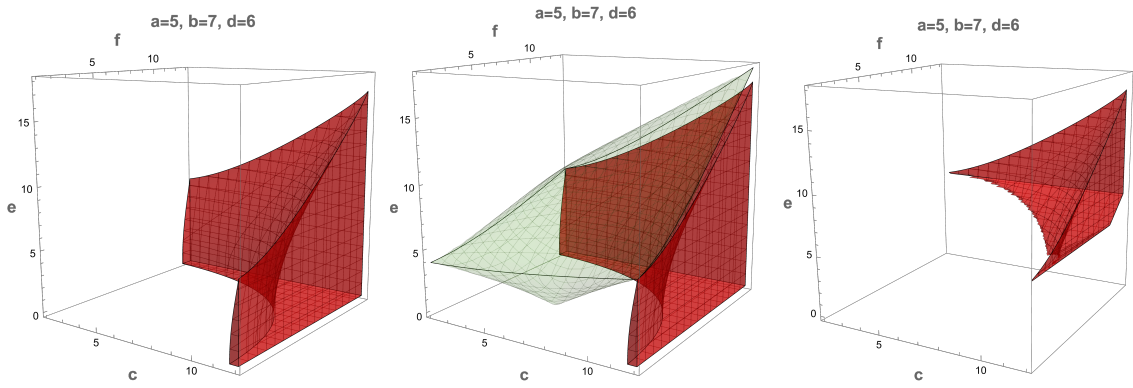


Figure 3: (Left) The region for which (c, e, f) violates Eq. (11) with $(a, b, c) = (5, 7, 6)$. (Middle) Comparing the region of tetrahedral (c, e, f) with those which violate Eq. (11). (Right) The values of (c, e, f) satisfying Eq. (8) but violating Eq. (11). [fig:ssa_extra_mileage](#)

2.4 Comparison with the CSW inequality

At this stage, it is worth comparing the inequality in Eq. (34) to a similar inequality derived by Christandl, Şahinoğlu, and Walter [CSW18], which holds for tetrahedral eigenvalues $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ which are non-negative and sufficiently small in the sense that $\lambda_1 + |\lambda| \leq 1$:

$$h(|\gamma|) + |\gamma| S(\bar{\gamma}) + h(|\phi|) + |\phi| S(\bar{\phi}) - h(|\beta|) - |\beta| S(\bar{\beta}) - h(|\epsilon|) - |\epsilon| S(\bar{\epsilon}) \geq 0. \quad (\text{eq:SSA_like_CSW}) \quad (12)$$

The inequality appearing in Eq. (34) is equivalent to the following by Eq. (110):

$$-|\gamma| \log |\gamma| + |\gamma| S(\bar{\gamma}) - |\phi| \log |\phi| + |\phi| S(\bar{\phi}) + |\beta| \log |\beta| - |\beta| S(\bar{\beta}) + |\epsilon| \log |\epsilon| - |\epsilon| S(\bar{\epsilon}) \geq 0. \quad (\text{eq:SSA_like_ineq_equiv}) \quad (13)$$

The difference between Eq. (12) and Eq. (13) is captured by a formal substitution replacing the binary entropy $h(|\lambda|)$ with the expression $-|\lambda| \log |\lambda|$. We note that it is possible to derive Eq. (13) from Eq. (12) by a downscaling argument. Specifically, while the set of tetrahedral eigenvalues is closed under scaling all eigenvalues by a non-negative r ,

$$(\alpha, \beta, \gamma, \delta, \epsilon, \phi) \mapsto (r\alpha, r\beta, r\gamma, r\delta, r\epsilon, r\phi), \quad (\text{eq:tetrahedral_scaling}) \quad (14)$$

the form and positivity of Eq. (12) is *not* invariant under scaling by r . Define q_r by the formula

$$q_r := r^{-1} (h(r|\gamma|) + h(r|\phi|) - h(r|\beta|) - h(r|\epsilon|)). \quad (15)$$

Letting $r \in (0, 1)$ be positive and small, we see that

$$r^{-1} h(r|\lambda|) = |\lambda| (1 - \log r) - |\lambda| \log |\lambda| + O(r^2) \quad (16)$$

which implies the following limit holds when $|\gamma| + |\phi| = |\beta| + |\epsilon|$:

$$q_0 := \lim_{r \rightarrow 0} q_r = -|\gamma| \log |\gamma| - |\phi| \log |\phi| + |\beta| \log |\beta| + |\epsilon| \log |\epsilon|. \quad (17)$$

In summary, by scaling down the eigenvalues in Eq. (12) by $r \in (0, 1)$ according to Eq. (14) and then scaling up Eq. (12) by multiplication by r^{-1} and taking the limit as $r \rightarrow \infty$, one recovers Eq. (13). Moreover, it can be shown that $q_r \geq q_0$ for all r and thus Eq. (13) is strictly stronger than Eq. (12). To prove this, we see that the trace equalities imply

$$\begin{aligned} q_r - q_0 &= r^{-1} (1 - r|\beta|) \log(1 - r|\beta|) + r^{-1} (1 - r|\epsilon|) \log(1 - r|\epsilon|) \\ &\quad - r^{-1} (1 - r|\gamma|) \log(1 - r|\gamma|) - r^{-1} (1 - r|\phi|) \log(1 - r|\phi|). \end{aligned} \quad (18)$$

Approximating $-(1-x) \log(1-x)$ by a Taylor series for $x \approx 0$,

$$-(1-x) \log(1-x) = x - \frac{x^2}{2} + O(x^3), \quad (19)$$

produces

$$q_r - q_0 = \frac{r}{2} (|\beta|^2 + |\epsilon|^2 - |\gamma|^2 - |\phi|^2) + O(r^2) = r |\alpha| |\delta| + O(r^2) \quad (20)$$

which implies $q_r > q_0$ and least for sufficiently small r . To prove $q_r > q_0$ holds for all r , by continuity, it therefore suffices to show that the gradient of $q_r - q_0$ with respect to r never changes sign for $r > 0$.

$$\begin{aligned} \partial_r (q_r - q_0) &= \frac{1}{r^2} (\log(1 - r|\gamma|) + \log(1 - r|\phi|) - \log(1 - r|\beta|) - \log(1 - r|\epsilon|)) \\ &= \frac{1}{r^2} \log \left(\frac{(1 - r|\gamma|)(1 - r|\phi|)}{(1 - r|\beta|)(1 - r|\epsilon|)} \right). \end{aligned} \quad (21)$$

Thus the gradient $\partial_r (q_r - q_0)$ vanishes only if the argument of the above logarithm is one, or equivalently when

$$0 = (1 - r|\gamma|)(1 - r|\phi|) - (1 - r|\beta|)(1 - r|\epsilon|) = r^2 (|\gamma| |\phi| - |\beta| |\epsilon|) = r^2 |\alpha| |\delta|. \quad (22)$$

As we are implicitly assuming $|\alpha| \neq 0$ and $|\delta| \neq 0$, we see that the gradient of $q_r - q_0$ never changes when $r > 0$ and thus $q_r - q_0$ is increasing with increasing $r > 0$.

3 Submodular and supermodular inequalities on positive semidefinite matrices

[lem:quadratic_ineq](#)

Lemma 2. *Let $A, B, C \geq 0$ be positive semidefinite Hermitian matrices, then*

$$\sum_j \lambda_j(A+B)^2 + \sum_j \lambda_j(B+C)^2 \leq \sum_j \lambda_j(B)^2 + \sum_j \lambda_j(A+B+C)^2. \quad (23)$$

Proof. Note that $\sum_j \lambda_j(X)^2 = \text{Tr}[X^2]$ for any Hermitian operator X . Therefore,

$$\begin{aligned} & \text{Tr}[(B)^2] + \text{Tr}[(A+B+C)^2] - \text{Tr}[(A+B)^2] - \text{Tr}[(B+C)^2] \\ &= (\text{Tr}[B^2]) + (\text{Tr}[A^2] + \text{Tr}[B^2] + \text{Tr}[C^2] + 2\text{Tr}[AB] + 2\text{Tr}[BC] + 2\text{Tr}[AC]) \\ &\quad - (\text{Tr}[A^2] + \text{Tr}[B^2] + 2\text{Tr}[AB]) - (\text{Tr}[B^2] + \text{Tr}[C^2] + 2\text{Tr}[BC]) \\ &= 2\text{Tr}[AC]. \end{aligned} \quad (24)$$

Since $A, C \geq 0$ are both positive semidefinite, we also have $\text{Tr}[AC] \geq 0$ which proves the claim. $\square$

Bouhrou, Gaubert, and Sagnol claim to prove the following family of inequalities [\[BGS08, Lemma 5.3\]](#). [lem:mme_operator_ineq](#)

Lemma 3. *Let $A, B, C \geq 0$ be positive semidefinite Hermitian matrices and let $p \in [0, 1]$. Then*

$$\sum_j \lambda_j(A+B)^p + \sum_j \lambda_j(B+C)^p \geq \sum_j \lambda_j(B)^p + \sum_j \lambda_j(A+B+C)^p. \quad (25)$$

The following operator inequality can be used to prove a number of inequalities [\[TCL11, Lemma 2.2\]](#). [lem:mme_operator_ineq](#)

Lemma 4. *Let $A, B, C \geq 0$ be positive semidefinite matrices and let k be a positive integer. Then*

$$(A+B+C)^{\otimes k} + B^{\otimes k} \geq (A+B)^{\otimes k} + (B+C)^{\otimes k}. \quad (26)$$

Proof. The result is obtained from applying a non-commutative multinomial expansion and noting that every summand on the right-hand-side above also appears on the left-hand-side above. $\square$

To obtain an inequality which depends only on the eigenvalues of $B, A+B, B+C$, and $A+B+C$, one can simply pair the positive semidefinite operator in Lemma 4 with an $\text{SU}(d)$ -equivariant and positive semidefinite operator:

[cor:isotypic_ineq](#)

Corollary 5. *Let $A, B, C \geq 0$ be positive semidefinite matrices and let $\lambda \vdash k$ be a Young diagram with k boxes. Then*

$$\text{Tr}[P_\lambda(A+B+C)^{\otimes k}] + \text{Tr}[P_\lambda B^{\otimes k}] \geq \text{Tr}[P_\lambda(A+B)^{\otimes k}] + \text{Tr}[P_\lambda(B+C)^{\otimes k}]. \quad (27)$$

Paksoy, Turkmen, and Zhang proved a family of polynomial inequalities which are generally not functions of just the eigenvalues [\[PTZ14\]](#). These inequalities are in terms a so-called generalized matrix function first defined by Marcus and Minc [\[MM65\]](#).

Definition 1 (Generalized matrix function). Let H be a subgroup of S_n and let χ be an irreducible character of H . The *generalized matrix function* for the character χ of H is

$$d_\chi^H(X) = \sum_{\sigma \in H} \chi(\sigma) \prod_i X_{i, \sigma(i)} \quad (28)$$

The notion of a *generalized matrix function* generalizes the notion of an *immanant* (when $H = S_n$) which, in turn, contains as special cases the notion of a permanent and determinant [\[HM21\]](#). [thm:ptz_ineqs](#)

Theorem 6. *Let $A, B, C \geq 0$ be positive semidefinite $n \times n$ matrices and let H be a subgroup of S_n with irreducible character χ . Then*

$$d_\chi^H(A+B+C) + d_\chi^H(B) \geq d_\chi^H(A+B) + d_\chi^H(B+C). \quad (29)$$

Proof. See [PTZ14]. □

As a corollary of Theorem 6, we obtain the following determinant and permanent inequalities.

Corollary 7. *Let $A, B, C \geq 0$ be positive semidefinite matrices. Then*

$$\det(A + B + C) + \det(B) \geq \det(A + B) + \det(B + C), \quad (30)$$

$$\text{per}(A + B + C) + \text{per}(B) \geq \text{per}(A + B) + \text{per}(B + C). \quad (31)$$

Note that with the exception of the above determinant inequality, the inequalities provided by Theorem 6 are generally not equivalent to inequalities involving only eigenvalues.

One also has another determinant inequality which is non-linear in determinants [TCL11, Lemma 2.3]. [lem:det_prod_ineq](#)

Lemma 8. *Let $A, B, C \geq 0$ be positive semidefinite matrices. Then*

$$\det(A + B + C) \det(B) \leq \det(A + B) \det(B + C). \quad (32)$$

Note that Lemma 8 can be seen as a special of Lemma 3 in the limit as $p \rightarrow 0$; the derivative of Lemma 3 in p at $p = 0$ yields

$$\sum_j \log \lambda_j(A + B) + \sum_j \log \lambda_j(B + C) \geq \sum_j \log \lambda_j(B) + \sum_j \log \lambda_j(A + B + C), \quad (33)$$
[{eq:bouhtou_ineq_p_zero}](#)

which is equivalent to Lemma 8. [thm:ssa_ineq](#)

Theorem 9. *Let $A, B, C \geq 0$ be positive semidefinite Hermitian matrices, then*

$$\sum_j \lambda_j(A + B) \log \lambda_j(A + B) + \sum_j \lambda_j(B + C) \log \lambda_j(B + C) \quad (34)$$
[{eq:ssa_like_ineq}](#)

$$\leq \sum_j \lambda_j(B) \log \lambda_j(B) + \sum_j \lambda_j(A + B + C) \log \lambda_j(A + B + C). \quad (35)$$

The most comprehensive result concerning the eigenvalues of partial sums of Hermitian matrices we have encountered thus far is [Sag13, Proposition 2.3] which provides the following result.

Theorem 10. *Let $A, B, C \geq 0$ be positive semidefinite Hermitian matrices and let f be a real valued and differentiable function over the non-negative real numbers $\mathbb{R}_{\geq 0}$ such that the derivative of f is an operator monotone function over the positive real numbers $\mathbb{R}_{> 0}$. Then the following inequality holds*

$$\text{Tr } f(A + B + C) + \text{Tr } f(B) \geq \text{Tr } f(A + B) + \text{Tr } f(B + C), \quad (36)$$

or equivalently,

$$\sum_j f(\lambda_j(A + B + C)) + \sum_j f(\lambda_j(B)) \geq \sum_j f(\lambda_j(A + B)) + \sum_j f(\lambda_j(B + C)) \quad (37)$$

Proof. The proof can be found in [Sag13, Section 2.1] □

As corollaries one obtains some of the previously mentioned inequalities for various functions f satisfying the conditions of the theorem; for example, choosing $f(x) = x \log x$ yields Theorem 9, while choosing $f(x) = -x^p$ for $p \in (0, 1]$ yields Lemma 3, and choosing $f(x) = x^p$ for $p \in (0, 2]$ yields the following.

Corollary 11. *Let $A, B, C \geq 0$ be positive semidefinite Hermitian matrices and let $p \in (0, 2]$. Then*

$$\sum_j \lambda_j(A + B)^p + \sum_j \lambda_j(B + C)^p \leq \sum_j \lambda_j(B)^p + \sum_j \lambda_j(A + B + C)^p. \quad (38)$$

4 Asymptotics of $SU(d)$ $6j$ operators

The purpose of this section is to derive some upper and lower bounds on generalized $6j$ symbol for $SU(d)$ in relation to the existence or non-existence of Hermitian operators satisfying the tetrahedral relations Eq. (1). This section essentially constitutes a generalization of the proof techniques in the “quantum-information-theoretic” approach taken in [Chr08]. Specifically, the results of this section are derived from considering a type of pseudo-commutative binomial expansion of tensor powers of sums of Hermitian operators and applying spectrum estimation to the resulting terms.

4.1 Schur-Weyl decomposition & spectrum estimation

Throughout this section, we make use of the following decomposition of a tensor-power of a d -dimensional complex vector space.

Proposition 12 (Schur-Weyl decomposition). *Let n and d be positive integers. Then the decomposition of $(\mathbb{C}^d)^{\otimes n}$ into irreducible representations of $S_n \times SU(d)$, known as the Schur-Weyl decomposition is given by*

$$(\mathbb{C}^d)^{\otimes n} \cong \bigoplus_{\lambda \vdash_n} W_\lambda \otimes V_\lambda^d \quad \text{\textcolor{blue}{(eq:schur_weyl_decomp)}} \quad (39)$$

where the direct sum is indexed by Young diagrams λ with n boxes and at most $d - 1$ rows.

The following result constrains the associated Schur-Weyl decomposition of the n -th tensor power, $X^{\otimes n}$, of a positive semidefinite Hermitian matrix X [Chr08; KW01].

Proposition 13 (Spectrum estimation). *Let P_λ denote the Hermitian projection operator associated to the isotypic subspace indexed by $\lambda \vdash_n$ in Eq. (39) and let X be a positive semidefinite matrix with eigenvalues χ . Then*

$$\text{Tr}(P_\lambda X^{\otimes n}) \leq |\chi|^n \dim(V_\lambda^d) \exp(-nH(\bar{\lambda}|\bar{\chi})), \quad \text{\textcolor{blue}{prop:spec_est}} \quad (40)$$

where $H(\bar{\lambda}|\bar{\chi})$ denotes the Kullback-Leibler divergence of $\bar{\lambda}$ from $\bar{\chi}$.

Proof. See [KW01] for a proof (noting that the $|\chi|^n$ term simply accounts for the possibility that $\text{Tr}(X) \neq 1$ because $\text{Tr}(X) = |\chi|$). $\square$

4.2 Schur-Weyl presentation of the $6j$ -operator

Lemma 14. *Let x, y, z arbitrary positive integers and let $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ be Young diagrams of the following sizes:*

$$\begin{array}{lll} \alpha \vdash_d x, & \beta \vdash_d y, & \gamma \vdash_d x + y, \\ \delta \vdash_d z, & \epsilon \vdash_d x + y + z, & \phi \vdash_d y + z, \end{array} \quad \text{\textcolor{blue}{(eq:xyz_sized_young_diagrams)}} \quad (41)$$

and define corresponding projection operators acting on $(\mathbb{C}^d)^{\otimes x+y+z}$ by

$$\begin{array}{lll} \Pi_\alpha := P_\alpha \otimes \text{id}^{\otimes y} \otimes \text{id}^{\otimes z}, & \Pi_\beta := \text{id}^{\otimes x} \otimes P_\beta \otimes \text{id}^{\otimes z}, & \Pi_\gamma := P_\gamma \otimes \text{id}^{\otimes z}, \\ \Pi_\delta := \text{id}^{\otimes x} \otimes \text{id}^{\otimes y} \otimes P_\delta, & \Pi_\epsilon := P_\epsilon, & \Pi_\phi := \text{id}^{\otimes x} \otimes P_\phi. \end{array} \quad \text{\textcolor{blue}{(eq:extended_projs)}} \quad (42)$$

Then every pair of projection operators in the set $\{\Pi_\alpha, \Pi_\beta, \Pi_\gamma, \Pi_\delta, \Pi_\epsilon, \Pi_\phi\}$ is commuting with only possible exception being Π_γ and Π_ϕ . Then $\Pi_{\alpha\beta\delta\epsilon}^\gamma$ and $\Pi_{\alpha\beta\delta\epsilon}^\phi$, defined by

$$\begin{aligned} \Pi_{\alpha\beta\delta\epsilon}^\gamma &= \Pi_\gamma \Pi_\alpha \Pi_\beta \Pi_\delta \Pi_\epsilon, \\ \Pi_{\alpha\beta\delta\epsilon}^\phi &= \Pi_\phi \Pi_\alpha \Pi_\beta \Pi_\delta \Pi_\epsilon. \end{aligned} \quad \text{\textcolor{blue}{(eq:commuting_extended_projs)}} \quad (43)$$

are projection operators. Moreover, then $6j$ -operator $\left\{ \begin{smallmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{smallmatrix} \right\}$ has operator norm satisfying

$$\left\| \left\{ \begin{smallmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{smallmatrix} \right\} \right\|_\infty = \left\| \Pi_{\alpha\beta\delta\epsilon}^\phi \Pi_{\alpha\beta\delta\epsilon}^\gamma \right\|_\infty. \quad \text{\textcolor{blue}{(eq:ud_sixj_prod_proj_operator_norm)}} \quad (44)$$

and Hilbert-Schmidt norm satisfying

$$\left\| \begin{Bmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{Bmatrix} \right\|_{\text{HS}}^2 = \frac{\text{Tr}(\Pi_{\alpha\beta\delta\epsilon}^\phi \Pi_{\alpha\beta\delta\epsilon}^\gamma)}{\dim(V_\epsilon^d) \dim(W_\alpha) \dim(W_\beta) \dim(W_\delta)}. \quad \text{\textcolor{blue}{(eq:ud_sixj_prod_proj_hs_norm)}} \quad (45)$$

Proof. That all projection operators except the pair (Π_γ, Π_ϕ) commute follows from elementary applications of Schur's lemma; the obstruction for the pair (Π_γ, Π_ϕ) is simply that Π_γ is not necessarily $\text{SU}(d)$ -equivariant with respect to the diagonal action on $\mathbb{C}^{\otimes y+z}$ and likewise Π_ϕ is not necessarily $\text{SU}(d)$ -equivariant with respect to the diagonal on $\mathbb{C}^{\otimes x+y}$. To prove both Eq. (44) and Eq. (45), it suffices to show there exists appropriately-typed isometries ι_I and ι_{II} such that

$$\Pi_{\alpha\beta\delta\epsilon}^\phi \Pi_{\alpha\beta\delta\epsilon}^\gamma = \iota_{II} \left(\text{id}_{V_\epsilon^d} \otimes \text{id}_{W_\alpha} \otimes \text{id}_{W_\beta} \otimes \text{id}_{W_\delta} \otimes \begin{Bmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{Bmatrix} \right) \iota_I^\dagger \quad \text{\textcolor{blue}{(eq:ud_sixj_prod_proj_iso)}} \quad (46)$$

This follows from the follow diagrammatic calculation TODO: Explain this.:

□
defn:Q_quantity

Definition 2. Let (A, B, C, D, E, F) be a tetrahedral sextuple of non-zero positive semidefinite $d \times d$ Hermitian matrices with eigenvalues $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$. Let x, y, z be arbitrary positive integers and define

$$\Omega := \frac{A^{\otimes x} \otimes B^{\otimes y} \otimes D^{\otimes z}}{|\alpha|^x |\beta|^y |\delta|^z}. \quad (48)$$

Let $(\alpha', \beta', \gamma', \delta', \epsilon', \phi')$ be Young diagrams with sizes determined by Eq. (41), and let $\Pi_{\alpha'\beta'\delta'\epsilon'}^{\gamma'}$ and $\Pi_{\alpha'\beta'\delta'\epsilon'}^{\phi'}$ be defined according to Eq. (43). Define $Q_{A,B,C,D,E,F}^{\alpha'\beta'\gamma'\delta'\epsilon'\phi'} \in \mathbb{C}$ by

$$Q_{A,B,C,D,E,F}^{\alpha'\beta'\gamma'\delta'\epsilon'\phi'} := \text{Tr}(\Pi_{\alpha'\beta'\delta'\epsilon'}^{\phi'} \Omega \Pi_{\alpha'\beta'\delta'\epsilon'}^{\gamma'}). \quad (49)$$

Proposition 15. Let everything be as in Definition 2. Then

$$Q_{A,B,C,D,E,F}^{\alpha'\beta'\gamma'\delta'\epsilon'\phi'} = \text{Tr}(\Pi_{\alpha'} \Pi_{\beta'} \Pi_{\gamma'} \Pi_{\delta'} \Pi_{\epsilon'} \Pi_{\phi'} \Omega), \quad \text{\textcolor{blue}{(eq:Q_alt)}} \quad (50)$$

and therefore

$$1 = \text{Tr}(\Omega) = \sum_{\alpha' \vdash_d x} \sum_{\beta' \vdash_d y} \sum_{\gamma' \vdash_d x+y} \sum_{\delta' \vdash_d z} \sum_{\epsilon' \vdash_d y+z} \sum_{\phi' \vdash_d x+y+z} Q_{A,B,C,D,E,F}^{\alpha' \beta' \gamma' \delta' \epsilon' \phi'} \quad \text{\textcolor{blue}{\textit{eq:completeness_relation}}} \quad (51)$$

Moreover, the magnitude of $Q_{A,B,C,D,E,F}^{\alpha' \beta' \gamma' \delta' \epsilon' \phi'}$ can be upper bounded by either

$$\left| Q_{A,B,C,D,E,F}^{\alpha' \beta' \gamma' \delta' \epsilon' \phi'} \right| \leq (\text{Tr}(\Pi_{\alpha'} \Omega) \text{Tr}(\Pi_{\beta'} \Omega) \text{Tr}(\Pi_{\gamma'} \Omega) \text{Tr}(\Pi_{\delta'} \Omega) \text{Tr}(\Pi_{\epsilon'} \Omega) \text{Tr}(\Pi_{\phi'} \Omega))^{\frac{1}{6}}. \quad \text{\textcolor{blue}{\textit{eq:Q_pre_upper_bound}}} \quad (52)$$

or

$$\left| Q_{A,B,C,D,E,F}^{\alpha' \beta' \gamma' \delta' \epsilon' \phi'} \right| \leq \left\| \Pi_{\alpha' \beta' \delta' \epsilon'}^{\phi'} \Pi_{\alpha' \beta' \delta' \epsilon'}^{\gamma'} \right\|_{\infty}. \quad \text{\textcolor{blue}{\textit{eq:Q_sixj_upper_bound}}} \quad (53)$$

Proof. The proof of Eq. (50) and Eq. (51) follows from elementary properties of the projection operators defined in Lemma 14. Moreover, Eq. (53) follows because Ω is positive semidefinite and trace one. To prove Eq. (52), consider an application of the Cauchy-Schwarz inequality to the Hilbert-Schmidt inner product, $\langle X, Y \rangle = \text{Tr}(X^\dagger Y)$, whenever $X = \Omega^{\frac{1}{2}} \Pi_1$ and $Y = \Omega^{\frac{1}{2}} \Pi_2$ (where Π_1 and Π_2 are generic Hermitian projectors and Ω is positive semidefinite):

$$\begin{aligned} |\text{Tr}(\Pi_1 \Omega \Pi_2)|^2 &= \left| \text{Tr}((\Omega^{\frac{1}{2}} \Pi_1)^\dagger (\Omega^{\frac{1}{2}} \Pi_2)) \right|^2 = \left| \langle \Omega^{\frac{1}{2}} \Pi_1, \Omega^{\frac{1}{2}} \Pi_2 \rangle \right|^2 \\ &\leq \langle \Omega^{\frac{1}{2}} \Pi_1, \Omega^{\frac{1}{2}} \Pi_1 \rangle \langle \Omega^{\frac{1}{2}} \Pi_2, \Omega^{\frac{1}{2}} \Pi_2 \rangle = \text{Tr}(\Pi_1 \Omega) \text{Tr}(\Pi_2 \Omega) \end{aligned} \quad \text{\textcolor{blue}{\textit{eq:biproj_ineq}}} \quad (54)$$

Letting $\Pi_1 = \Pi_{\phi'} \Pi_{\alpha'} \Pi_{\beta'}$ and $\Pi_2 = \Pi_{\gamma'} \Pi_{\delta'} \Pi_{\epsilon'}$ and applying Eq. (54) yields

$$\begin{aligned} \left| Q_{A,B,C,D,E,F}^{\alpha' \beta' \gamma' \delta' \epsilon' \phi'} \right|^6 &= |\text{Tr}[(\Pi_{\phi'} \Pi_{\alpha'} \Pi_{\beta'}) \Omega (\Pi_{\gamma'} \Pi_{\delta'} \Pi_{\epsilon'})]|^6 \\ &\leq \text{Tr}(\Pi_{\phi'} \Pi_{\alpha'} \Pi_{\beta'} \Omega)^3 \text{Tr}(\Pi_{\gamma'} \Pi_{\delta'} \Pi_{\epsilon'} \Omega)^3 \\ &\leq \text{Tr}(\Pi_{\phi'} \Omega) \text{Tr}(\Pi_{\alpha'} \Omega) \text{Tr}(\Pi_{\beta'} \Omega) \text{Tr}(\Pi_{\gamma'} \Omega) \text{Tr}(\Pi_{\delta'} \Omega) \text{Tr}(\Pi_{\epsilon'} \Omega). \end{aligned} \quad (55)$$

\text{\textcolor{blue}{\textit{thm:Q_upper_bound}}} \quad \square

Theorem 16. Let everything be as in Definition 2 except fix x, y, z to be

$$x := \lceil n |\alpha| \rceil, \quad y := \lceil n |\beta| \rceil, \quad z := \lceil n |\delta| \rceil, \quad (56)$$

where n is a sufficiently large positive integer ensuring that x, y and z are non-zero. Then

$$\left| Q_{A,B,C,D,E,F}^{\alpha' \beta' \gamma' \delta' \epsilon' \phi'} \right| \leq p(n) \exp(-n I_{\alpha \beta \gamma \delta \epsilon \phi}^{\alpha' \beta' \gamma' \delta' \epsilon' \phi'} / 6). \quad \text{\textcolor{blue}{\textit{eq:Q_upper_bound}}} \quad (57)$$

where $p(n)$ is a polynomial in n and where

$$I_{\alpha \beta \gamma \delta \epsilon \phi}^{\alpha' \beta' \gamma' \delta' \epsilon' \phi'} = |\alpha| H(\bar{\alpha}' | \bar{\alpha}) + |\beta| H(\bar{\beta}' | \bar{\beta}) + |\gamma| H(\bar{\gamma}' | \bar{\gamma}) + |\delta| H(\bar{\delta}' | \bar{\delta}) + |\epsilon| H(\bar{\epsilon}' | \bar{\epsilon}) + |\phi| H(\bar{\phi}' | \bar{\phi}). \quad (58)$$

TODO: Is this relative entropy deviation measure conceptually useful or is it best to relax the deviation measure to its geometric analogue using Pinsker's inequality?

Proof. To begin the proof, we consider x, y and z to be arbitrary positive integers. For each $\lambda' \in \{\alpha', \beta', \gamma', \delta', \epsilon', \phi'\}$, it is possible to upper-bound $\text{Tr}(\Pi_{\lambda'} \Omega)$ using the spectrum estimation technique Proposition 13. For a detailed example, we will consider the bound on $\text{Tr}(\Pi_{\gamma'} \Omega)$.

$$\text{Tr}(\Pi_{\gamma'} \Omega) = \frac{\text{Tr}[(P_{\gamma'} \otimes \text{id}^{\otimes z})(A^{\otimes x} \otimes B^{\otimes y} \otimes D^{\otimes z})]}{|\alpha|^x |\beta|^y |\delta|^z} = \frac{\text{Tr}[P_{\gamma'}(A^{\otimes x} \otimes B^{\otimes y})] |\delta|^z}{|\alpha|^x |\beta|^y |\delta|^z}. \quad (59)$$

Using the assumption that $C = A + B$, we have $C^{\otimes x+y} = (A + B)^{\otimes x+y}$ which implies

$$\text{Tr}(P_{\gamma'} C^{\otimes x+y}) = \sum_{k=0}^{x+y} \binom{x+y}{k} \text{Tr}(P_{\gamma'} (A^{\otimes k} \otimes B^{\otimes x+y-k})). \quad (60)$$

Here we have used the S_{x+y} -equivariance of $P_{\gamma'}$. Since each of the summands above are non-negative, we conclude for all $0 \leq k \leq x+y$ that

$$\mathrm{Tr}(P_{\gamma'} C^{\otimes x+y}) \geq \binom{x+y}{k} \mathrm{Tr}(P_{\gamma'} (A^{\otimes k} \otimes B^{\otimes x+y-k})). \quad (61)$$

Letting $k = x$ and rearranging yields

$$\mathrm{Tr}(P_{\gamma'} (A^{\otimes x} \otimes B^{\otimes y})) \leq \frac{x!y!}{(x+y)!} \mathrm{Tr}(P_{\gamma'} C^{\otimes x+y}). \quad (62)$$

Furthermore, spectrum estimation (Proposition 13) yields

$$\mathrm{Tr}(\Pi_{\gamma'} \Omega) \leq \frac{1}{|\alpha|^x |\beta|^y} \frac{x!y!}{(x+y)!} \mathrm{Tr}[P_{\gamma'} C^{\otimes x+y}], \quad (63)$$

$$\leq \frac{(|\alpha| + |\beta|)^{x+y}}{|\alpha|^x |\beta|^y} \frac{x!y!}{(x+y)!} \dim(V_{\gamma'}^d) \exp(-(x+y)H(\bar{\gamma}'|\bar{\gamma})). \quad (64)$$

Analogous calculations for $\lambda' \in \{\alpha', \beta', \gamma', \delta', \epsilon', \phi'\}$ produce the following set of upper bounds:

$$\mathrm{Tr}(\Pi_{\alpha'} \Omega) \leq \dim(V_{\alpha'}^d) \exp(-xH(\bar{\alpha}'|\bar{\alpha})), \quad (65)$$

$$\mathrm{Tr}(\Pi_{\beta'} \Omega) \leq \dim(V_{\beta'}^d) \exp(-yH(\bar{\beta}'|\bar{\beta})), \quad (66)$$

$$\mathrm{Tr}(\Pi_{\gamma'} \Omega) \leq \frac{(|\alpha| + |\beta|)^{x+y}}{|\alpha|^x |\beta|^y} \frac{x!y!}{(x+y)!} \dim(V_{\gamma'}^d) \exp(-(x+y)H(\bar{\gamma}'|\bar{\gamma})), \quad (67)$$

$$\mathrm{Tr}(\Pi_{\delta'} \Omega) \leq \dim(V_{\delta'}^d) \exp(-zH(\bar{\delta}'|\bar{\delta})), \quad (68)$$

$$\mathrm{Tr}(\Pi_{\epsilon'} \Omega) \leq \frac{(|\alpha| + |\beta| + |\delta|)^{x+y+z}}{|\alpha|^x |\beta|^y |\delta|^z} \frac{x!y!z!}{(x+y+z)!} \dim(V_{\epsilon'}^d) \exp(-(x+y+z)H(\bar{\epsilon}'|\bar{\epsilon})), \quad (69)$$

$$\mathrm{Tr}(\Pi_{\phi'} \Omega) \leq \frac{(|\beta| + |\delta|)^{x+y+z}}{|\beta|^y |\delta|^z} \frac{y!z!}{(y+z)!} \dim(V_{\phi'}^d) \exp(-(y+z)H(\bar{\phi}'|\bar{\phi})). \quad (70)$$

Our objective now will be to use the given values for x, y, z to prove the multinomial prefactors appearing in the bounds for γ', ϵ' , and ϕ' reasonably small compared to n . An application of Lemma 25 (when $k = 2$) implies the following bound

$$\frac{(|\alpha| + |\beta|)^{x+y}}{|\alpha|^x |\beta|^y} \frac{x!y!}{(x+y)!} \leq e^4 (n(|\alpha| + |\beta|) + 3)^2. \quad (71)$$

Moreover, using [CM06, Equation 7], we have

$$\dim(V_{\gamma'}^d) \leq (|\gamma'| + 1)^{d(d-1)/2} = (x+y+1)^{d(d-1)/2} < (n(|\alpha| + |\beta|) + 3)^{d(d-1)/2}. \quad (72)$$

Altogether, we can relax the bound on $\mathrm{Tr}(\Pi_{\gamma'} \Omega)$ to obtain

$$\mathrm{Tr}(\Pi_{\gamma'} \Omega) \leq e^4 (n|\epsilon| + 3)^{\frac{d(d-1)}{2}+2} \exp(-n|\gamma| H(\bar{\gamma}'|\bar{\gamma})). \quad (73)$$

Similar calculations for $\mathrm{Tr}(\Pi_{\epsilon'} \Omega)$ yields (noting that $k = 3$ when applying Lemma 25):

$$\mathrm{Tr}(\Pi_{\epsilon'} \Omega) \leq e^{16} (n|\epsilon| + 4)^{\frac{d(d-1)}{2}+3} \exp(-n|\epsilon| H(\bar{\epsilon}'|\bar{\epsilon})). \quad (74)$$

It is also not too difficult to see that the polynomial in n term associated to the upper bound on $\mathrm{Tr}(\Pi_{\epsilon'} \Omega)$ above is the largest such subexponential term and thus Eq. (57) indeed holds if we take $p(n)$ to be

$$p(n) = e^{16} (n|\epsilon| + 4)^{\frac{d(d-1)}{2}+3}. \quad (75)$$

□

Corollary 17. Let $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ be the eigenvalues of non-zero positive semidefinite tetrahedral Hermitian matrices (A, B, C, D, E, F) . Then there exists a sextuple of sequences of Young diagrams $(\alpha'_n, \beta'_n, \gamma'_n, \delta'_n, \epsilon'_n, \phi'_n)$ of sizes

$$\begin{aligned} \alpha'_n \vdash_d [n|\alpha|], & \quad \beta'_n \vdash_d [n|\beta|], & \quad \gamma'_n \vdash_d [n|\alpha|] + [n|\beta|], \\ \delta'_n \vdash_d [n|\delta|], & \quad \epsilon'_n \vdash_d [n|\alpha|] + [n|\beta|] + [n|\delta|], & \quad \phi'_n \vdash_d [n|\beta|] + [n|\delta|], \end{aligned} \quad (76)$$

satisfying (i) the convergence conditions

$$\begin{aligned} \lim_{n \rightarrow \infty} \bar{\alpha}'_n &= \bar{\alpha}, & \lim_{n \rightarrow \infty} \bar{\beta}'_n &= \bar{\beta}, & \lim_{n \rightarrow \infty} \bar{\gamma}'_n &= \bar{\gamma}, \\ \lim_{n \rightarrow \infty} \bar{\delta}'_n &= \bar{\delta}, & \lim_{n \rightarrow \infty} \bar{\epsilon}'_n &= \bar{\epsilon}, & \lim_{n \rightarrow \infty} \bar{\phi}'_n &= \bar{\phi}, \end{aligned} \quad (77)$$

and (ii)

$$\left\| \begin{Bmatrix} \alpha'_n & \beta'_n & \gamma'_n \\ \delta'_n & \epsilon'_n & \phi'_n \end{Bmatrix} \right\|_{\infty} \geq \frac{1}{(n|\epsilon| + 4)^{6(d-1)}}. \quad (78)$$

Proof. For each n , it suffices to pick $(\alpha'_n, \beta'_n, \gamma'_n, \delta'_n, \epsilon'_n, \phi'_n)$ be the sextuple which maximizes the value of $|Q_{A,B,C,D,E,F}^{\alpha'_n, \beta'_n, \gamma'_n, \delta'_n, \epsilon'_n, \phi'_n}|$. Such a choice will necessarily satisfy a lower bound due to the completeness relation Eq. (51). Specifically,

$$\left| Q_{A,B,C,D,E,F}^{\alpha'_n, \beta'_n, \gamma'_n, \delta'_n, \epsilon'_n, \phi'_n} \right| \geq \frac{1}{q(n)} \quad (79)$$

where $q(n)$ is any upper bound on the number of possible summands in Eq. (51) viewed as a function of n . As there are at most $(k+1)^{d-1}$ Young diagrams with $d-1$ rows and size at most k , letting

$$q(n) = (n|\epsilon| + 4)^{6(d-1)} \quad (80)$$

ensures Eq. (79) holds because the size of ϵ'_n is at most $k = n|\epsilon| + 3$. Moreover, applying Eq. (57) yields

$$I_{\alpha\beta\gamma\delta\epsilon\phi}^{\alpha'_n, \beta'_n, \gamma'_n, \delta'_n, \epsilon'_n, \phi'_n} \leq \frac{6}{n} (\log q(n) + \log p(n)) \xrightarrow{n \rightarrow \infty} 0 \quad (81)$$

and thus Eq. (77) holds because $H(\bar{\lambda}'|\bar{\lambda}) \geq 0$ with equality if and only if $\bar{\lambda}' = \bar{\lambda}$. Then Eq. (78) follows from Eq. (79) combined with Eq. (53) and Eq. (44). $\square$

Theorem 18. Let $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ be non-negative non-zero eigenvalues which are not tetrahedral. Then for all sequences of Young diagrams $(\alpha'_n, \beta'_n, \gamma'_n, \delta'_n, \epsilon'_n, \phi'_n)$ satisfying Eq. (76) and Eq. (77), we have

$$\left\| \begin{Bmatrix} \alpha'_n & \beta'_n & \gamma'_n \\ \delta'_n & \epsilon'_n & \phi'_n \end{Bmatrix} \right\|_{\text{HS}}^2 \leq \exp(-nI_*^{\alpha\beta\gamma\delta\epsilon\phi}/6 + O(\log n)). \quad (82)$$

where $I_*^{\alpha\beta\gamma\delta\epsilon\phi} > 0$ quantifies the deviation of $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ from being tetrahedral:

$$I_*^{\alpha\beta\gamma\delta\epsilon\phi} = \min_{\substack{\tilde{\alpha}\tilde{\beta}\tilde{\gamma}\tilde{\delta}\tilde{\epsilon}\tilde{\phi} \\ \text{tetrahedral}}} I_{\tilde{\alpha}\tilde{\beta}\tilde{\gamma}\tilde{\delta}\tilde{\epsilon}\tilde{\phi}}^{\alpha\beta\gamma\delta\epsilon\phi}. \quad (83)$$

Proof. Let dX be the normalized Hilbert-Schmidt measure over $d \times d$ density matrices and let k be a positive integer. Then define an operator Γ_k on $(\mathbb{C}^d)^{\otimes k}$ by

$$\Gamma_k := \int dX X^{\otimes k}. \quad (84)$$

As Γ_k is known to be both S_k and $\text{SU}(d)$ equivariant, by Schur's lemma, Γ_k can be expressed as

$$\Gamma_k = \sum_{\lambda' \vdash_d k} r_{\lambda'} P_{\lambda'} \quad (85)$$

with coefficients $r_{\lambda'}$ that can be computed [ZS01; Wal14] to obtain

$$r_{\lambda'} = \frac{\dim(V_{\lambda'}^d)}{\dim(W_{\lambda'}) \dim(V_{(k)}^{d^2})}. \quad (86)$$

Now let $(\alpha', \beta', \gamma', \delta', \epsilon', \phi')$ be arbitrary Young diagrams with respective sizes $(x, y, x+y, z, x+y+z, y+z)$ where x, y, z are determined by a positive integer n as in Theorem 16. Using Eq. (45), we obtain the formula

$$\left\| \begin{Bmatrix} \alpha' & \beta' & \gamma' \\ \delta' & \epsilon' & \phi' \end{Bmatrix} \right\|_{\text{HS}}^2 = \frac{\text{Tr}(\Pi_{\alpha'\beta'\delta'\epsilon'}^{\phi'} \Pi_{\alpha'\beta'\delta'\epsilon'}^{\gamma'})}{\dim(V_{\epsilon'}^d) \dim(W_{\alpha'}) \dim(W_{\beta'}) \dim(W_{\delta'})} \quad (87)$$

$$= R_{\alpha', \beta', \delta', \epsilon'} \text{Tr}(\Pi_{\alpha'\beta'\delta'\epsilon'}^{\phi'} (\Gamma_x \otimes \Gamma_y \otimes \Gamma_z) \Pi_{\alpha'\beta'\delta'\epsilon'}^{\gamma'}) \quad (88)$$

where the coefficient $R_{\alpha', \beta', \delta', \epsilon'}$ is a ratio of dimensions:

$$R_{\alpha', \beta', \delta', \epsilon'} := \frac{\dim(V_{(x)}^{d^2}) \dim(V_{(y)}^{d^2}) \dim(V_{(z)}^{d^2})}{\dim(V_{\epsilon'}^d) \dim(V_{\alpha'}^d) \dim(V_{\beta'}^d) \dim(V_{\delta'}^d)}. \quad (89)$$

Moreover, by the definition of Γ_k , we have

$$\text{Tr}(\Pi_{\alpha'\beta'\delta'\epsilon'}^{\phi'} (\Gamma_x \otimes \Gamma_y \otimes \Gamma_z) \Pi_{\alpha'\beta'\delta'\epsilon'}^{\gamma'}) \quad (90)$$

$$= \int dX \int dY \int dZ \text{Tr}(\Pi_{\alpha'\beta'\delta'\epsilon'}^{\phi'} (X^{\otimes x} \otimes Y^{\otimes y} \otimes Z^{\otimes z}) \Pi_{\alpha'\beta'\delta'\epsilon'}^{\gamma'}), \quad (91)$$

where the integral is taken over a triple of $d \times d$ density matrices X, Y, Z . For each integrand, define a tetrahedral sextuple $(\tilde{A}, \tilde{B}, \tilde{C}, \tilde{D}, \tilde{E}, \tilde{F})$ as follows

$$\begin{aligned} \tilde{A} &:= |\alpha| X, & \tilde{B} &:= |\beta| Y, & \tilde{C} &:= |\alpha| X + |\beta| Y, \\ \tilde{D} &:= |\delta| Z, & \tilde{E} &:= |\alpha| X + |\beta| Y + |\gamma| Z, & \tilde{F} &:= |\beta| Y + |\delta| Z. \end{aligned} \quad (92)$$

Furthermore, let $(\tilde{\alpha}, \tilde{\beta}, \tilde{\gamma}, \tilde{\delta}, \tilde{\epsilon}, \tilde{\phi})$ be the eigenvalues of $(\tilde{A}, \tilde{B}, \tilde{C}, \tilde{D}, \tilde{E}, \tilde{F})$. In this notation, the integrand above is equal to

$$Q_{\tilde{A}, \tilde{B}, \tilde{C}, \tilde{D}, \tilde{E}, \tilde{F}}^{\alpha' \beta' \gamma' \delta' \epsilon' \phi'} = \text{Tr}(\Pi_{\alpha'\beta'\delta'\epsilon'}^{\phi'} (X^{\otimes x} \otimes Y^{\otimes y} \otimes Z^{\otimes z}) \Pi_{\alpha'\beta'\delta'\epsilon'}^{\gamma'}) \quad (93)$$

Moreover, by Eq. (53) we have

$$\left| Q_{\tilde{A}, \tilde{B}, \tilde{C}, \tilde{D}, \tilde{E}, \tilde{F}}^{\alpha' \beta' \gamma' \delta' \epsilon' \phi'} \right| \leq p(n) \exp(-n I_{\tilde{\alpha} \tilde{\beta} \tilde{\gamma} \tilde{\delta} \tilde{\epsilon} \tilde{\phi}}^{\alpha' \beta' \gamma' \delta' \epsilon' \phi'} / 6). \quad (94)$$

Under the assumption that $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ are not tetrahedral, we conclude that

$$(\alpha, \beta, \gamma, \delta, \epsilon, \phi) \neq (\tilde{\alpha}, \tilde{\beta}, \tilde{\gamma}, \tilde{\delta}, \tilde{\epsilon}, \tilde{\phi}) \quad (95)$$

and therefore $I_{\tilde{\alpha} \tilde{\beta} \tilde{\gamma} \tilde{\delta} \tilde{\epsilon} \tilde{\phi}}^{\alpha \beta \gamma \delta \epsilon \phi} > 0$. **TODO: Figure out how to describe the convergence, is there a triangle inequality for relative entropies? Maybe we need to relax the rate function I to its geometric analogue using Pinsker's inequality, then everything works.** $\square$

4.3 The asymptotics of irrep dimensions

It will be useful for what follows to better understand how the dimension, $\dim(V_{\lambda}^d)$, of the irreducible representation V_{λ}^d of $\text{SL}(d)$ grows as d becomes much larger $|\lambda|$ while $|\lambda|$ itself is much larger than $\ell(\lambda)$. Itzykson and Nauenberg provide the following estimate for $\dim(V_{\lambda}^d)$ [IN66].

Lemma 19. *Let λ be an positive integer partition of length $\ell := \ell(\lambda)$, let $d \geq \ell$ be a positive integer and let $\rho^d := (d-1, d-2, \dots, 2, 1)$ be the Weyl-vector for $\text{SL}(d)$. The following approximation holds when $d \gg |\lambda|^2$ and $\lambda_{\ell} \gg \ell$:*

$$\dim(V_{\lambda}^d) = \dim(V_{\lambda}^{\ell}) \times \prod_{i=1}^{\ell} \frac{\rho_i^{\ell}!}{(\lambda_i + \rho_i^{\ell})!} \times d^{|\lambda|} \times \left(1 + O\left(\frac{|\lambda|^2}{d}\right) \right). \quad (96)$$

Proof. Weyl's character formula can be used to derive the following formula for $\dim(V_\lambda^d)$ in terms of the Young diagram λ and the dimension d [IN66]:

$$\dim(V_\lambda^d) = \frac{\Delta(\lambda + \rho^d)}{\Delta(\rho^d)} = \prod_{1 \leq i < j \leq d} \frac{(\lambda_i + \rho_i^d) - (\lambda_j + \rho_j^d)}{\rho_i^d - \rho_j^d} = \prod_{1 \leq i < j \leq d} \frac{(\lambda_i - i) - (\lambda_j - j)}{j - i}, \quad (97)$$

where $\Delta(x) = \prod_{i < j} (x_i - x_j)$ is the Vandermonde determinant. Isolating the terms above which depend on d from those that do not yields

$$\dim(V_\lambda^d) = \dim(V_\lambda^\ell) \times \prod_{i=1}^{\ell} \frac{\rho_i^\ell!}{(\lambda_i + \rho_i^\ell)!} \times \prod_{i=1}^{\ell} \frac{(\lambda_i + \rho_i^d)!}{\rho_i^d!}. \quad \text{\textcolor{blue}{\code{eq:dim_separation_large_d}}} \quad (98)$$

Given positive integers a and b such that $b \gg a^2$, one obtains the following approximation:

$$\begin{aligned} \log \left(\frac{(a+b)!}{b!} \right) &= \log \prod_{j=1}^a (b+j) = \sum_{j=1}^a \log(b+j) \\ &= a \log b + \sum_{j=1}^a \log \left(1 + \frac{j}{b} \right) = a \log b + O \left(\frac{a^2}{b} \right). \end{aligned} \quad (99)$$

The assumption that $d \gg |\lambda|^2$ and $i \leq \ell \ll \lambda_\ell \leq |\lambda|$ also implies $d - i = \rho_i^d \gg \lambda_i^2$. Therefore,

$$\log \left(\frac{(\lambda_i + \rho_i^d)!}{\rho_i^d!} \right) = \lambda_i \log \rho_i^d + O \left(\frac{\lambda_i^2}{\rho_i^d} \right). \quad (100)$$

The assumptions also imply the following simplification

$$\log \left(\frac{(\lambda_i + \rho_i^d)!}{\rho_i^d!} \right) = \lambda_i \log d + O \left(\frac{|\lambda|^2}{d} \right) \quad (101)$$

which, when applied to Eq. (98), proves the claim. $\square$

Using Lemma 19, it can be seen that for fixed λ , $\dim(V_\lambda^d)$ generally grows polynomially in d with leading exponent $|\lambda|$ and leading coefficient depending only on λ . The objective of the following lemma is to characterize the asymptotics of this leading coefficient for large $|\lambda|$ and even large d . TODO: Figure out a elegant way to present these “double asymptotics”, maybe just epsilonfying everything, like $\max\{\ell/|\lambda|, |\lambda|/d\} \leq \epsilon$. \code{lem:double_asymptotics}

Lemma 20. *Let everything be as in Lemma 19. Then*

$$\log \dim(V_\lambda^d) = |\lambda| + |\lambda| \log(d) + s(\lambda) + O(\ell^2 \log |\lambda|) + O \left(\frac{|\lambda|^2}{d} \right) \quad (102)$$

where $s(\lambda)$ is defined by

$$s(\lambda) := - \sum_{i=1}^{\ell(\lambda)} \lambda_i \log \lambda_i. \quad \text{\textcolor{blue}{\code{eq:unnorm_shannon}}} \quad (103)$$

Proof. The proof proceeds by considering the multiplicands in Eq. (96) separately. First, we upper bound the dimension $\dim(V_\lambda^\ell)$; for indices $i < j \leq \ell$, one has $(\lambda_i - i) - (\lambda_j - j) \leq |\lambda| + 1$ as well as $j - i \geq 1$ which implies

$$\dim(V_\lambda^\ell) = \prod_{1 \leq i < j \leq \ell} \frac{(\lambda_i - i) - (\lambda_j - j)}{j - i} \leq (|\lambda| + 1)^{\ell(\ell-1)/2}, \quad \text{\textcolor{blue}{\code{eq:dim_lambda_ell_bound}}} \quad (104)$$

and thus $\log \dim(V_\lambda^\ell) = O(\ell^2 \log |\lambda|)$.

Second, we apply Stirling's approximation [Rob55] to the second term in Eq. (96) under the given assumption that $\lambda_\ell \gg \ell$ (which implies $\rho_i^\ell \ll \lambda_i$),

$$\log \left(\prod_{i=1}^{\ell} \frac{\rho_i^\ell!}{(\lambda_i + \rho_i^\ell)!} \right) = \sum_{i=1}^{\ell} \log \rho_i^\ell! - \sum_{i=1}^{\ell} \log(\lambda_i + \rho_i^\ell)!, \quad (105)$$

$$= |\lambda| + \sum_{i=1}^{\ell} \left(\rho_i^\ell \log \rho_i^\ell - (\lambda_i + \rho_i^\ell) \log(\lambda_i + \rho_i^\ell) + O(\log \lambda_i) \right), \quad (106)$$

$$= |\lambda| - \sum_{i=1}^{\ell} \lambda_i \log \lambda_i + O(\ell \log |\lambda|), \quad (107)$$

$$\text{\textcolor{blue}{(eq:unnorm_shannon_term)}} \quad (108)$$

Combining Eq. (104) and Eq. (108) with Lemma 19 proves the claim. $\square$

Remark 2. Note that the quantity $s(\lambda)$ defined in Eq. (103) resembles the Shannon entropy,

$$S(p) = - \sum_{i=1}^{\ell} p_i \log p_i \quad (109)$$

of a probability distribution $p = (p_1, \dots, p_\ell)$. Indeed, $s(\lambda)$ is related to the Shannon entropy of the *normalization* of λ by the formula

$$s(\lambda) = |\lambda| S \left(\frac{\lambda}{|\lambda|} \right) - |\lambda| \log |\lambda|. \quad \text{\textcolor{blue}{(eq:little_s_relation)}} \quad (110)$$

TODO: Perhaps a more direct way of relating the asymptotics of $\dim(V_\lambda^d)$ to $\dim(W_\lambda)$ can be found in Harrow's paper [Har23, Lemma 2]:

Lemma 21. *Let λ be a Young diagram with at most d parts. Then*

$$\frac{|\lambda|!}{d^{|\lambda|}} \frac{\dim(V_\lambda^d)}{\dim(W_\lambda)} = \prod_{(i,j) \in \lambda} \left(1 + \frac{j-i}{d} \right) \quad (111)$$

4.4 An entropic inequality for tetrahedral spectra

thm:ud_recouple_bound

Theorem 22. *Let $(\alpha, \beta, \gamma, \delta, \epsilon, \phi) \in \mathbb{Z}_{\text{dec}}^d$ be a sextuple of dominant weights for $\text{GL}(d)$ and let $\left\{ \begin{smallmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{smallmatrix} \right\}$ be the corresponding $\text{GL}(d)$ -recoupling operator. Then*

$$\left\| \left\{ \begin{smallmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{smallmatrix} \right\} \right\|_{HS}^2 \leq \left(c_{\alpha, \phi}^\epsilon c_{\beta, \delta}^\phi c_{\alpha, \beta}^\gamma c_{\gamma, \delta}^\epsilon \right)^{\frac{1}{2}} \frac{\dim(V_\gamma) \dim(V_\phi)}{\dim(V_\beta) \dim(V_\epsilon)} \quad (112)$$

Proof. **TODO:** Explain what this all means carefully. Applying both the Cauchy-Schwarz inequality with

respect to the contraction annotated below, and Schur's lemma multiple times yields the following inequality:

$$\begin{aligned}
& \left(\epsilon \right)^2 C_{\alpha,\beta}^\gamma C_{\alpha,\phi}^\epsilon C_{\beta,\delta}^\phi C_{\gamma,\delta}^\epsilon = \\
& \left[\begin{array}{c} \left\{ \begin{array}{ccc} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{array} \right\}^\dagger \\ \left\{ \begin{array}{ccc} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{array} \right\} \end{array} \right] = \\
& \left[\begin{array}{c} \text{Diagram 1} \\ \text{Diagram 2} \end{array} \right] = \left[\begin{array}{c} \text{Diagram 3} \\ \text{Diagram 4} \end{array} \right] \\
& \left[\begin{array}{c} \text{Diagram 5} \\ \text{Diagram 6} \end{array} \right]^{\frac{1}{2}} = \frac{1}{\left(\beta \right) \left(\epsilon \right)} \left[\begin{array}{c} \text{Diagram 7} \\ \text{Diagram 8} \end{array} \right]^{\frac{1}{2}} \\
& = \frac{\left(\epsilon \right)^2 \left(\gamma \right) \left(\phi \right)}{\left(\beta \right) \left(\epsilon \right)} \left[\begin{array}{c} \text{Diagram 9} \\ \text{Diagram 10} \\ \text{Diagram 11} \\ \text{Diagram 12} \end{array} \right]^{\frac{1}{2}}
\end{aligned}
\tag{113}$$

lem:asymptotic_dimension_ratio □

Lemma 23. *Let $\gamma, \phi, \beta, \epsilon$ be a quadruple of Young diagrams such that*

$$|\gamma| + |\phi| = |\beta| + |\epsilon| \tag{114}$$

and that $|\epsilon| = \max\{|\gamma|, |\phi|, |\beta|, |\epsilon|\}$. Let $\ell := \max\{\ell(\gamma), \ell(\phi), \ell(\beta), \ell(\epsilon)\}$ and let d be a positive integer such that $d \geq \ell$. Then

$$\log \left(\frac{\dim(V_\gamma^d) \dim(V_\phi^d)}{\dim(V_\beta^d) \dim(V_\epsilon^d)} \right) = I(\gamma, \phi; \beta, \epsilon) + O(\ell^2 \log |\epsilon|) + O\left(\frac{\text{lex-asymptotic-dimension-ratio}}{d}\right) \quad (115)$$

where $I(\gamma, \phi; \beta, \epsilon)$ is the function

$$I(\gamma, \phi; \beta, \epsilon) := s(\gamma) + s(\phi) - s(\beta) - s(\epsilon). \quad (116)$$

Moreover, $I(k\gamma, k\phi; k\beta, k\epsilon) = kI(\gamma, \phi; \beta, \epsilon)$ for any $k > 0$.

Proof. The proof follows from four applications of Lemma 20. The assumption that the sizes of the Young diagrams are balanced in the sense of Eq. (114) implies that the $\log(d)$ terms appearing in Lemma 20 cancel each other and one obtains Eq. (115).

Four applications of Eq. (110) yields

$$\begin{aligned}
I(\gamma, \phi; \beta, \epsilon) &= s(\gamma) + s(\phi) - s(\beta) - s(\epsilon) \\
&= \{|\gamma| S(\bar{\gamma}) + |\phi| S(\bar{\phi}) - |\beta| S(\bar{\beta}) - |\epsilon| S(\bar{\epsilon})\} + \dots \\
&\quad \dots - \{|\gamma| \log |\gamma| + |\phi| \log |\phi| - |\beta| \log |\beta| - |\epsilon| \log |\epsilon|\}.
\end{aligned} \tag{117}$$

Under scaling $\lambda \mapsto k\lambda$ for any $k > 0$, observe that $\overline{k\lambda} = \bar{\lambda}$ and $|k\lambda| = k|\lambda|$. Therefore the above expression implies

$$I(k\gamma, k\phi; k\beta, k\epsilon) = kI(\gamma, \phi; \beta, \epsilon) + (|\gamma| + |\phi| - |\beta| - |\epsilon|)k \log k. \tag{118}$$

Since Eq. (114) is assumed to hold, the coefficient in front of the $k \log k$ term appearing in Eq. (118) vanishes and therefore $I(k\gamma, k\phi; k\beta, k\epsilon) = kI(\gamma, \phi; \beta, \epsilon)$ as claimed. $\square$

Theorem 24. *Let $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ be the eigenvalues of Hermitian matrices (A, B, C, D, E, F) satisfying Eq. (1). Then necessarily $|\beta| + |\epsilon| = |\gamma| + |\phi|$ and furthermore*

$$I(\gamma, \phi; \beta, \epsilon) \geq 0. \tag{119}$$

Proof. The proof follows from asymptotics given by Lemma 23 in the limit as $d \rightarrow \infty$, the upper bound provided by Theorem 22, and the lower bound provided by Corollary 17. **TODO: Finish this** $\square$

5 Conclusion

References

- [AB68] Vishnu K Agrawala and Johan G Belinfante. “Graphical formulation of recoupling theory for any compact group”. In: *Annals of Physics* 49.1 (1968), pp. 130–170.
- [AH11] Daiki Akimoto and Masahito Hayashi. “Discrimination of the change point in a quantum setting”. In: *Physical Review A* 83.5 (2011), p. 052328.
- [AHP10] Koenraad Audenaert, Fumio Hiai, and Dénes Petz. “Strongly subadditive functions”. In: *Acta Mathematica Hungarica* 128.4 (2010), pp. 386–394.
- [Ais97] AK Aiston. “A skein theoretic proof of the hook formula for quantum dimension”. In: *arXiv preprint q-alg/9711019* (1997).
- [Alc+23] Judith Alcock-Zeilinger et al. “Wigner 6j symbols for SU (N): Symbols with at least two quark-lines”. In: *Journal of Mathematical Physics* 64.2 (2023).
- [Alc18] Judith M Alcock-Zeilinger. *The Special Unitary Group, Birdtracks, and Applications in QCD*. 2018.
- [Ale+18] Anton Alekseev et al. “Concentration of symplectic volumes on Poisson homogeneous spaces”. In: *arXiv preprint arXiv:1808.06975* (2018).
- [Ale97] A Yu Alekseev. “On Poisson actions of compact Lie groups on symplectic manifolds”. In: *Journal of Differential Geometry* 45.2 (1997), pp. 241–256.
- [AM07] Anton Alekseev and Eckhard Meinrenken. “Ginzburg-Weinstein via Gelfand-Zeitlin”. In: *Journal of Differential Geometry* 76.1 (2007), pp. 1–34.
- [AM98] AK Aiston and HR Morton. “Idempotents of Hecke algebras of type A”. In: *Journal of Knot Theory and Its Ramifications* 7.04 (1998), pp. 463–487.
- [AMS20] Victor Alekseev, Andrey Morozov, and Alexey Sleptsov. “Multiplicity-free $U_q(\mathfrak{sl}_N)$ 6-j symbols: relations, asymptotics, symmetries”. In: *Nuclear Physics B* 960 (2020), p. 115164.
- [AMW00] Anton Alekseev, Eckhard Meinrenken, and Christopher Woodward. “Group-valued equivariant localization”. In: *Inventiones mathematicae* 140.2 (2000), pp. 327–350.
- [AMW01] Anton Alekseev, Eckhard Meinrenken, and Chris Woodward. “Linearization of Poisson actions and singular values of matrix products”. In: *Annales de l’institut Fourier*. Vol. 51. 6. 2001, pp. 1691–1717.
- [AMW99] Anton Alekseev, Eckhard Meinrenken, and Chris Woodward. “Duistermaat-Heckman distributions for group valued moment maps”. In: *arXiv preprint math/9903087* (1999).
- [APS17a] Anton Alekseev, Masha Podkopaeva, and Andras Szenes. “A symplectic proof of the Horn inequalities”. In: *Advances in Mathematics* 318 (2017), pp. 711–736.
- [APS17b] Anton Alekseev, Masha Podkopaeva, and Andras Szenes. “The Horn problem and planar networks”. In: *Advances in Mathematics* 318 (2017), pp. 678–710.
- [Aqu+12] Vincenzo Aquilanti et al. “Semiclassical mechanics of the Wigner 6j-symbol”. In: *Journal of Physics A: Mathematical and Theoretical* 45.6 (2012), p. 065209.
- [Art23] DV Artamonov. “Classical-symbols of finite-dimensional representations of the algebra”. In: *Theoretical and Mathematical Physics* 216.1 (2023), pp. 909–923.
- [Art24] Dmitry Artamonov. “Calculation of 6j-symbols for the Lie algebra $\mathfrak{gl}_n$ ”. In: *arXiv preprint arXiv:2405.05628* (2024).
- [AW97] Sharad Agnihotri and Chris Woodward. “Eigenvalues of products of unitary matrices and quantum Schubert calculus”. In: *arXiv preprint alg-geom/9712013* (1997).
- [Bac10] Miriam Backens. “The 6j symbols and an extended version of Horn’s problem”. In: (2010).
- [Bar03] John W Barrett. “Geometrical measurements in three-dimensional quantum gravity”. In: *International Journal of Modern Physics A* 18.suppl02 (2003), pp. 97–113.
- [BDS11] Eugenio Bianchi, Pietro Dona, and Simone Speziale. “Polyhedra in loop quantum gravity”. In: *Physical Review D* 83.4 (2011), p. 044035. arXiv: [1009.3402v2](#).

- [Bel02] Prakash Belkale. “Geometric proofs of Horn and saturation conjectures”. In: *arXiv preprint math/0208107* (2002).
- [Bel04] Prakash Belkale. “Invariant theory of $GL(n)$ and intersection theory of Grassmannians”. In: *International Mathematics Research Notices* 2004.69 (2004), pp. 3709–3721.
- [Bel08] Prakash Belkale. “Quantum generalization of the Horn conjecture”. In: *Journal of the American Mathematical Society* 21.2 (2008), pp. 365–408.
- [BGS08] Mustapha Bouhtou, Stéphane Gaubert, and Guillaume Sagnol. “Optimization of network traffic measurement: a semidefinite programming approach”. In: *Proceedings of the International*. Vol. 230. 2008.
- [Bha01] Rajendra Bhatia. “Linear algebra to quantum cohomology: the story of Alfred Horn’s inequalities”. In: *The American Mathematical Monthly* 108.4 (2001), pp. 289–318.
- [BK16] Prakash Belkale and Shrawan Kumar. “The multiplicative eigenvalue problem and deformed quantum cohomology”. In: *Advances in Mathematics* 288 (2016), pp. 1309–1359.
- [BN23] Bruce Bartlett and Nzaganya Nzaganya. “Coherent loop states and angular momentum”. In: *arXiv preprint arXiv:2306.17293* (2023).
- [Boa07] Philip P Boalch. “Regge and Okamoto symmetries”. In: *Communications in mathematical physics* 276.1 (2007), pp. 117–130.
- [BVW18] Nicole Berline, Michèle Vergne, and Michael Walter. “The Horn inequalities from a geometric point of view”. In: *L’Enseignement Mathématique* 63.3 (2018), pp. 403–470.
- [Cha10] Laurent Charles. “On the quantization of polygon spaces”. In: (2010).
- [Chr08] Matthias Christandl. “A Quantum Information-Theoretic Proof of the Relation between Horn’s Problem and the Littlewood-Richardson Coefficients”. In: *Lecture Notes in Computer Science*. Springer Berlin Heidelberg, 2008, pp. 120–128.
- [CM06] Matthias Christandl and Graeme Mitchison. “The spectra of quantum states and the Kronecker coefficients of the symmetric group”. In: *Communications in mathematical physics* 261.3 (2006), pp. 789–797.
- [CMZ19] Robert Coquereaux, Colin McSwiggen, and Jean-Bernard Zuber. “Revisiting Horn’s problem”. In: *Journal of Statistical Mechanics: Theory and Experiment* 2019.9 (2019), p. 094018.
- [CMZ20] Robert Coquereaux, Colin McSwiggen, and Jean-Bernard Zuber. “On Horn’s problem and its volume function”. In: *Communications in Mathematical Physics* 376.3 (2020), pp. 2409–2439.
- [Coq23] Robert Coquereaux. “About integer-valued variants of the theta and $6j$ symbols”. In: *Journal of Mathematical Physics* 64.3 (2023).
- [CP95] Vyjayanthi Chari and Andrew N Pressley. *A guide to quantum groups*. Cambridge university press, 1995.
- [CŞW18] Matthias Christandl, M Burak Şahinoğlu, and Michael Walter. “Recoupling coefficients and quantum entropies”. In: *Annales Henri Poincaré*. Vol. 19. 2. Springer. 2018, pp. 385–410.
- [CW06] Nicola Ciccoli and Pawel Witkowski. “From Poisson to quantum geometry”. In: *Lecture Notes*, available at <http://toknotes.mimuw.edu.pl> (2006).
- [CZ18] Robert Coquereaux and Jean-Bernard Zuber. “From orbital measures to Littlewood–Richardson coefficients and hive polytopes”. In: *Annales de l’Institut Henri Poincaré D* 5.3 (2018), pp. 339–386.
- [DH05] Sumit Daftuar and Patrick Hayden. “Quantum state transformations and the Schubert calculus”. In: *Annals of Physics* 315.1 (2005), pp. 80–122.
- [DZ10] Amir Dembo and Ofer Zeitouni. *Large deviations techniques and applications*. Berlin: Springer, 2010.
- [Esp12] Chiara Esposito. “On the classical and quantum momentum map”. In: (2012). 1203.4112: [arxiv](https://arxiv.org/abs/1203.4112).

- [Feg78] HD Fegan. “The heat equation on a compact Lie group”. In: *Transactions of the American Mathematical Society* 246 (1978), pp. 339–357.
- [FG13] S Friedland and S Gaubert. “Submodular spectral functions of principal submatrices of a hermitian matrix, extensions and applications”. In: *Linear Algebra and its Applications* 438.10 (2013), pp. 3872–3884.
- [FK00] Laurent Freidel and Kirill Krasnov. “Simple spin networks as Feynman graphs”. In: *Journal of Mathematical Physics* 41.4 (2000), pp. 1681–1690.
- [FKL10] Laurent Freidel, Kirill Krasnov, and Etera R Livine. “Holomorphic factorization for a quantum tetrahedron”. In: *Communications in Mathematical Physics* 297.1 (2010), pp. 45–93.
- [FL03] Laurent Freidel and David Louapre. “Asymptotics of 6j and 10j symbols”. In: *Classical and Quantum Gravity* 20.7 (2003), p. 1267.
- [FNR07] L Freidel, K Noui, and Ph Roche. “6J symbols duality relations”. In: *Journal of mathematical physics* 48.11 (2007).
- [FT14] Peter J Forrester and Colin J Thompson. “The Golden-Thompson inequality: Historical aspects and random matrix applications”. In: *Journal of Mathematical Physics* 55.2 (2014).
- [Ful00] William Fulton. “Eigenvalues, invariant factors, highest weights, and Schubert calculus”. In: *Bulletin of the American Mathematical Society* 37.3 (2000), pp. 209–249.
- [Ful97] William Fulton. *Young tableaux: with applications to representation theory and geometry*. 35. Cambridge University Press, 1997.
- [GSS19] Friedrich Götze, Holger Sambale, and Arthur Sinulis. “Higher order concentration for functions of weakly dependent random variables”. In: *Electronic Journal of Probability* 24 (2019), pp. 1–19.
- [Gur08] Razvan Gurau. “The Ponzano–Regge Asymptotic of the 6 j Symbol: An Elementary Proof”. In: *Annales Henri Poincaré*. Vol. 9. Springer. 2008, pp. 1413–1424.
- [Har23] Aram W Harrow. “Approximate orthogonality of permutation operators, with application to quantum information”. In: *Letters in Mathematical Physics* 114.1 (2023), p. 1.
- [Hes20] Sarah Tanja Hess. “Examination of the connection between the Horn Problem and the Lax Conjecture”. In: (2020).
- [HHR16] Hal M Haggard, Muxin Han, and Aldo Riello. “Encoding curved tetrahedra in face holonomies: phase space of shapes from group-valued moment maps”. In: *Annales Henri Poincaré*. Vol. 17. Springer. 2016, pp. 2001–2048.
- [HK05] Andre Henriques and Joel Kamnitzer. *Crystals and coboundary categories*. 2005. arXiv: [math/0406478 \[math.QA\]](#).
- [HK06] André Henriques and Joel Kamnitzer. “The octahedron recurrence and gl_n crystals”. In: *Advances in Mathematics* 206.1 (2006), pp. 211–249.
- [HM21] Felix Huber and Hans Maassen. “Matrix forms of immanant inequalities”. In: *arXiv preprint arXiv:2103.04317* (2021).
- [Hor62] Alfred Horn. “Eigenvalues of sums of Hermitian matrices.” In: (1962).
- [Hub20] Felix Huber. *Positive Maps and Matrix Contractions from the Symmetric Group*. 2020. arXiv: [2002.12887](#).
- [HW22] Felix Huber and Nikolai Wyderka. *Refuting spectral compatibility of quantum marginals*. 2022. arXiv: [2211.06349v2](#).
- [IN66] Claude Itzykson and Michael Nauenberg. “Unitary groups: Representations and decompositions”. In: *Reviews of Modern Physics* 38.1 (1966), p. 95.
- [Kly00] Alexander A Klyachko. “Random walks on symmetric spaces and inequalities for matrix spectra”. In: *Linear Algebra and its Applications* 319.1-3 (2000), pp. 37–59.
- [Knu00] Allen Knutson. “The symplectic and algebraic geometry of Horn’s problem”. In: *Linear Algebra and its Applications* 319.1-3 (2000), pp. 61–81.

- [KPS24] Stefan Keppeler, Simon Plätzer, and Malin Sjö Dahl. “Wigner 6j symbols with gluon lines: completing the set of 6j symbols required for color decomposition”. In: *Journal of High Energy Physics* 2024.5 (2024), pp. 1–25.
- [KR90] AN Kirillov and N Yu Reshetikhin. “Representations of the Algebra $U_q(\mathfrak{sl}(2))$, q-Orthogonal Polynomials and Invariants of Links”. In: *New developments in the theory of knots* 11 (1990), p. 202.
- [KTW03] Allen Knutson, Terence Tao, and Christopher T Woodward. “A positive proof of the Littlewood-Richardson rule using the octahedron recurrence”. In: *arXiv preprint math/0306274* (2003).
- [KW01] Michael Keyl and Reinhard F Werner. “Estimating the spectrum of a density operator”. In: *Physical Review A* 64.5 (2001), p. 052311.
- [Lam05] McKenzie Lamb. “A proof of the Thompson conjecture using Poisson geometry”. In: (2005).
- [LE03] Jiang-Hua Lu and Sam Evens. “Thompson’s conjecture for real semi-simple Lie groups”. In: (2003). arXiv: [math/0310098](https://arxiv.org/abs/math/0310098).
- [LS23] Elena Lanina and Alexey Sleptsov. “Tug-the-hook symmetry for quantum 6j-symbols”. In: *Physics Letters B* 845 (2023), p. 138138.
- [McS21] Colin McSwiggen. “The Harish-Chandra integral: an introduction with examples”. In: *L’Enseignement Mathématique* 67.3 (2021), pp. 229–299.
- [MM65] Marvin Marcus and Henryk Minc. “Generalized matrix functions”. In: *Transactions of the American Mathematical Society* 116 (1965), pp. 316–329.
- [MMS17] A Mironov, A Morozov, and A Sleptsov. “On 6 j-symbols for symmetric representations of $U_q(\mathfrak{su}(N))$ ”. In: *JETP Letters* 106 (2017), pp. 630–636.
- [Moa84] Daniel S Moak. “The q-analogue of Stirling’s formula”. In: *The Rocky Mountain Journal of Mathematics* (1984), pp. 403–413.
- [Mou84] J Moussouris. “Quantum models of space-time based on recoupling theory”. PhD thesis. University of Oxford, 1984.
- [MR05] Annalisa Marzuoli and Mario Rasetti. “Computing spin networks”. In: *Annals of Physics* 318.2 (2005), pp. 345–407.
- [MS18] Andrey Morozov and Alexey Sleptsov. “New Symmetries for the $U_q(\mathfrak{sl}(N))$ 6-j Symbols from the Eigenvalue Conjecture¹”. In: *JETP Letters* 108 (2018), pp. 697–704.
- [PR68] Giorgio Ponzano and Tullio Eugenio Regge. “Semiclassical limit of Racah coefficients”. In: *Spectroscopic and group theoretical methods in physics* (1968).
- [PTZ14] Vehbi Paksoy, Ramazan Turkmen, and Fuzhen Zhang. “Inequalities of generalized matrix functions via tensor products”. In: *The Electronic Journal of Linear Algebra* 27 (2014), pp. 332–341.
- [Res87] N Yu Reshetikhin. *Quantized universal enveloping algebras, the Yang-Baxter equation and invariants of links, 1*. Tech. rep. 1987.
- [Ric02] G Richardson. “The trigonometry of the tetrahedron”. In: *The Mathematical Gazette* 2.32 (1902), pp. 149–158.
- [Rob02] Justin Roberts. “Asymptotics and 6j-symbols”. In: *Geom. Topol. Monogr* 4 (2002), pp. 245–261.
- [Rob55] Herbert Robbins. “A remark on Stirling’s formula”. In: *The American mathematical monthly* 62.1 (1955), pp. 26–29.
- [Rob99] Justin Roberts. “Classical 6j-symbols and the tetrahedron”. In: *Geometry & Topology* 3.1 (1999), pp. 21–66.
- [RR86] David P Robbins and Howard Rumsey Jr. “Determinants and alternating sign matrices”. In: *Advances in Mathematics* 62.2 (1986), pp. 169–184.
- [RW00] Tullio Regge and Ruth M Williams. “Discrete structures in gravity”. In: *Journal of Mathematical Physics* 41.6 (2000), pp. 3964–3984.

- [Sag13] Guillaume Sagnol. “Approximation of a maximum-submodular-coverage problem involving spectral functions, with application to experimental designs”. In: *Discrete Applied Mathematics* 161.1-2 (2013), pp. 258–276.
- [SBT17] David Sutter, Mario Berta, and Marco Tomamichel. “[Multivariate trace inequalities](#)”. In: *Communications in Mathematical Physics* 352.1 (2017), pp. 37–58. arXiv: [1604.03023v3](#).
- [SJS18] Meera Sitharam, Audrey John, and Jessica Sidman, eds. *Handbook of geometric constraint systems principles*. Discrete Mathematics and Its Applications. New York, NY: Productivity Press, July 2018.
- [Spe05] David E Speyer. “Horn’s problem, Vinnikov curves, and the hive cone”. In: (2005).
- [TCL11] Lin Tie, Kai-Yuan Cai, and Yan Lin. “Rearrangement inequalities for Hermitian matrices”. In: *Linear algebra and its applications* 434.2 (2011), pp. 443–456.
- [TT73] RC Thompson and S Therianos. “The Eigenvalues and Singular Values of Matrix Sums and Products. VII (1)”. In: *Canadian Mathematical Bulletin* 16.4 (1973), pp. 561–569.
- [TW06] Yuka U Taylor and Christopher T Woodward. “6 j symbols for and non-Euclidean tetrahedra”. In: *Selecta Mathematica* 11.3-4 (2006), p. 539.
- [VMK88] Dmitrii Aleksandrovich Varshalovich, Anatolij Nikolaevič Moskalev, and Valerii Kel’manovich Khersonskii. *Quantum theory of angular momentum*. World Scientific, 1988.
- [Wal14] Michael Walter. *Multipartite quantum states and their marginals*. 2014. arXiv: [1410.6820v1](#).
- [ZS01] Karol Zyczkowski and Hans-Jürgen Sommers. “[Induced measures in the space of mixed quantum states](#)”. In: *Journal of Physics A: Mathematical and General* 34.35 (2001), p. 7111.
- [Zub18] Jean-Bernard Zuber. “Horn’s problem and Harish-Chandra’s integrals. Probability density functions”. In: *Annales de l’Institut Henri Poincaré D* 5.3 (2018), pp. 309–338.

A Entropic identities & inequalities

In this section, we recall a handful of identities and inequalities concerning multinomial coefficients and the multinomial distribution in the context of Sanov's theorem [DZ10, Chapter 2]. Let $q = (q_1, \dots, q_k)$ be a probability distribution such that for all $i \in \{1, \dots, k\}$, nq_i is an integer. Let $M_n(q)$ denote the following multinomial coefficient:

$$M_n(q) := \binom{n}{nq_1, \dots, nq_k} = \frac{n!}{(nq_1)! \dots (nq_k)!}, \quad \text{\texttt{eq:multinomial_coefficient}} \quad (120)$$

which can be bounded (see [DZ10, Lemma 2.1.8]) in terms of $H(q)$ by ,

$$(n+1)^{-k} e^{nH(q)} \leq M_n(q) \leq e^{nH(q)}, \quad \text{\texttt{eq:multinomial_coefficient_bound}} \quad (121)$$

where $H(q) = -\sum_{i=1}^k q_i \log q_i$ is the Shannon entropy of q . Additionally, we have for any probability distribution $p = (p_1, \dots, p_k)$,

$$p_1^{nq_1} \dots p_k^{nq_k} = e^{-n(H(q) + H(q|p))} \quad \text{\texttt{eq:multinomial_probability}} \quad (122)$$

where $H(q|p) = \sum_{i=1}^k q_i \log q_i/p_i$ denotes the Kullback-Leibler divergence or relative entropy of q from p with the convention that $H(q|p) = \infty$ if $q_i > 0$ whenever $p_i = 0$. Note that $H(q|p) = 0$ if and only if $p = q$. Combining Eq. (121) with Eq. (122) places bounds on the multinomial distribution

$$(n+1)^{-k} e^{-nH(q|p)} \leq \binom{n}{nq_1, \dots, nq_k} p_1^{nq_1} \dots p_k^{nq_k} \leq e^{-nH(q|p)} \quad (123)$$

If needed, it is possible to relax these bounds further using Pinsker's inequality for $H(q|p)$ in conjunction with its reverse (which only holds for $H(q|p) < \infty$) [GSS19]:

$$2d_{\text{TV}}^2(p, q) \leq H(q|p) \leq \frac{4}{p_{\min}} d_{\text{TV}}^2(p, q) \quad (124)$$

where

$$d_{\text{TV}}(p, q) = \max_{1 \leq j \leq k} |p_j - q_j| = \frac{1}{2} \sum_{j=1}^k |p_j - q_j| \quad \text{\texttt{eq:pinskera}} \quad (125)$$

is known as the *total variation distance* between p and q . Now for any distribution $p = (p_1, \dots, p_k)$ and any positive integer n , it is possible to find another distribution $p' = (p'_1, \dots, p'_k)$ such that (i) np'_i is an integer for all $i \in \{1, \dots, k\}$, (ii) $H(p'|p) < \infty$ is finite, and (iii) p' approximates p in the sense that for some constant $C \geq 0$,

$$d_{\text{TV}}(p, p') \leq \frac{C}{n}. \quad \text{\texttt{eq:tv_approx}} \quad (126)$$

Concretely, one can choose p' such that

$$np'_j := \text{nint} \left(n \sum_{i=1}^j p_i \right) - \text{nint} \left(n \sum_{i=1}^{j-1} p_i \right) \quad \text{\texttt{eq:nint_approx}} \quad (127)$$

where $\text{nint} : \mathbb{R} \rightarrow \mathbb{Z}$ is the function which rounds each real number to the nearest integer (with half integers always rounded up). Using this approximation, we have $|np_j - np'_j| \leq 1$ because each rounding operation in Eq. (127) introduces an error of at most 1/2 and thus Eq. (126) holds with $C = 1$. In any case, when Eq. (126) holds for some constant $C \geq 0$, Eq. (125) reveals $H(p'|p) \leq \frac{4C}{p_{\min} n^2}$ which yields a useful, albeit crude, lower bound for the multinomial distribution:

$$\binom{n}{np'_1, \dots, np'_k} p_1^{np'_1} \dots p_k^{np'_k} \geq \exp \left(-\frac{4C^2}{p_{\min} n} - k \log(n+1) \right). \quad \text{\texttt{lem:multinomial_dist_lower_bound}} \quad (128)$$

Lemma 25. Let $k \geq 2$ and let $(w_1, \dots, w_k) \in \mathbb{R}_{>0}^k$ have minimum $w_{\min} := \min\{w_1, \dots, w_k\} > 0$ and let n be a positive integer satisfying $n \geq w_{\min}^{-1}$ and define positive integers $(z_1, \dots, z_k) \in \mathbb{N}_{>0}^k$ by the equation

$$z_i := \lceil nw_i \rceil. \quad (129)$$

Let $Z = z_1 + \dots + z_k$ and $W = w_1 + \dots + w_k$ and defined probability distributions $\bar{z}$ and $\bar{w}$ by

$$\bar{z}_i = \frac{z_i}{Z}, \quad \text{and} \quad \bar{w}_i = \frac{w_i}{W}, \quad (130)$$

for all $i \in \{1, \dots, k\}$. Then

$$\frac{Z!}{z_1! \dots z_k!} \frac{w_1^{z_1} \dots w_k^{z_k}}{W^Z} \geq e^{-4(k-1)^2} (nW + k + 1)^{-k} \quad \text{\textcolor{blue}{\code{eq:multinomial_dist_lower_bound}}} \quad (131)$$

Proof. Let $\delta_j := z_j - nw_j$ denote the rounding surplus such that $0 \leq \delta_j < 1$. Note that for $n \gg k$, we have $Z \approx nW$ because

$$nW \leq Z < nW + k. \quad (132)$$

In this regime, we wish to show that $\bar{z}_i \approx \bar{w}_i$.

$$|\bar{z}_i - \bar{w}_i| = \left| \frac{z_i}{Z} - \frac{w_i}{W} \right| = \frac{|Wz_i - Zw_i|}{ZW}. \quad (133)$$

The numerator above can be bounded independently of n by,

$$|Wz_i - Zw_i| \quad (134)$$

$$= |(nw_i + \delta_i)(w_1 + \dots + w_k) - w_i(\delta_1 + nw_1 + \dots + \delta_k + nw_k)|, \quad (135)$$

$$= |\delta_i(w_1 + \dots + \hat{w}_i + \dots + w_k) - w_i(\delta_1 + \dots + \hat{\delta}_i + \dots + \delta_k)|, \quad (136)$$

$$\leq |\delta_i(w_1 + \dots + \hat{w}_i + \dots + w_k)| + |w_i(\delta_1 + \dots + \hat{\delta}_i + \dots + \delta_k)|, \quad (137)$$

$$\leq (w_1 + \dots + \hat{w}_i + \dots + w_k) + (k-1)(w_i), \quad (138)$$

$$\leq (k-1)(w_1 + \dots + w_k), \quad (139)$$

$$= (k-1)W. \quad (140)$$

Consequently,

$$d_{\text{TV}}(\bar{z}, \bar{w}) = \max_{1 \leq i \leq k} |\bar{z}_i - \bar{w}_i| \leq \frac{k-1}{Z} \quad (141)$$

and therefore

$$\frac{Z!}{z_1! \dots z_k!} \frac{w_1^{z_1} \dots w_k^{z_k}}{W^Z} \geq (Z+1)^{-k} \exp(-ZH(\bar{z}|\bar{w})), \quad (142)$$

$$\geq (Z+1)^{-k} \exp \left[-\frac{4ZW}{w_{\min}} \left(\frac{k-1}{Z} \right)^2 \right], \quad (143)$$

$$\geq (nW + k + 1)^{-k} \exp \left[-\frac{4(k-1)^2}{nw_{\min}} \right], \quad (144)$$

$$\geq (nW + k + 1)^{-k} \exp(-4(k-1)^2). \quad (145)$$

□

TODO: Figure out the best lower bound on the multinomial distribution to use here.

\textcolor{blue}{\code{lem:multinomial_dist_lower_bound_alt}}

Lemma 26. Let everything be as in Lemma 25. Then

$$\frac{Z!}{z_1! \dots z_k!} \frac{w_1^{z_1} \dots w_k^{z_k}}{W^Z} \geq \left(\frac{1}{20} \right)^k \left(\frac{W}{w_1 \dots w_k} \right)^{\frac{1}{2}} \frac{\text{\textcolor{blue}{\code{eq:multinomial_dist_lower_bound_alt}}}}{n^{\frac{3k-1}{2}}} \quad (146)$$

Proof. The proof follows from a version of Stirling's approximation due to Robbins [Rob55] which states

$$n! = \sqrt{2\pi n} n^{n+\frac{1}{2}} e^{-n} e^{r_n}, \quad \text{where} \quad \frac{1}{12n+1} \leq r_n \leq \frac{1}{12n}. \quad (147)$$

Therefore,

$$\begin{aligned} \frac{Z!}{z_1! \cdots z_k!} \frac{w_1^{z_1} \cdots w_k^{z_k}}{W^Z} &= \frac{Z!}{W^Z} \frac{w_1^{z_1}}{z_1!} \cdots \frac{w_k^{z_k}}{z_k!} \\ &= \left(\frac{1}{2\pi}\right)^{\frac{k-1}{2}} \left(\frac{W}{w_1 \cdots w_k}\right)^{\frac{1}{2}} \left(\frac{Z}{W}\right)^{Z+\frac{1}{2}} \left(\frac{w_1}{z_1}\right)^{z_1+\frac{1}{2}} \cdots \left(\frac{w_k}{z_k}\right)^{z_k+\frac{1}{2}} \exp\left(r_Z - \sum_i r_{z_i}\right). \end{aligned} \quad (148)$$

In this setting, we have assumed $nw_i \geq 1$ which implies

$$nw_i \leq z_i < nw_i + 1 \leq 2nw_i, \quad \text{and} \quad nW \leq Z < nW + k < 2nW. \quad (150)$$

Consequently¹,

$$\left(\frac{Z}{W}\right)^{Z+\frac{1}{2}} \geq n^{nW+\frac{1}{2}}, \quad \left(\frac{w_i}{z_i}\right)^{z_i+\frac{1}{2}} \geq \left(\frac{w_i}{nw_i+1}\right)^{nw_i+\frac{3}{2}} \geq \frac{2^{-\frac{5}{2}}}{n^{nw_i+\frac{3}{2}}}. \quad (151)$$

and the remainder term satisfies

$$r_Z - \sum_i r_{z_i} \geq \frac{1}{12Z+1} - \sum_i \frac{1}{12z_i} \geq -\frac{k}{12}. \quad (152)$$

Altogether, we have

$$\frac{Z!}{z_1! \cdots z_k!} \frac{w_1^{z_1} \cdots w_k^{z_k}}{W^Z} = \left(\frac{1}{2\pi}\right)^{\frac{k-1}{2}} \left(\frac{W}{w_1 \cdots w_k}\right)^{\frac{1}{2}} \frac{2^{-\frac{5k}{2}}}{n^{\frac{3k-1}{2}}} \exp\left(-\frac{k}{12}\right). \quad (153)$$

Moreover,

$$\left(\frac{1}{2\pi}\right)^{\frac{k-1}{2}} 2^{-\frac{5k}{2}} \exp\left(-\frac{k}{12}\right) \geq \left(\frac{1}{20}\right)^k \quad (154)$$

□

¹Here we use the identity that $x \geq 1$ implies $\left(\frac{x}{x+1}\right)^{x+\frac{3}{2}} \geq 2^{-5/2}$

B [Research Notes]

B.1 Horn’s problem & generalizations

- Horn’s original paper where they conjectured the form of a necessary and sufficient set of inequalities via an inductive scheme for increasing dimension [Hor62]
- An excellent expository overview of the big picture by Knutson [Knu00]
- An excellent introduction to these ideas is the recent masters thesis on various aspects of additive and multiplicative Horn problems in connection to Vinnikov curves and the Lax conjecture due to Hess and supervised by S. Kuhlmann and M. Infusino [Hes20]
- Quantum-info style proof of Littlewood-Richardson coefficient connection due to Christandl [Chr08]
- Coquereaux and Zuber explicitly outline the link between the orbital measures (i.e. the induced distributions in the sense of Klyachko’s random walk approach) and the magnitude of the stretched Littlewood-Richardson coefficients [CZ18]. See also the papers [CMZ20; CMZ19; Zub18].
- A recent article by McSwiggen is an excellent introduction to the Harish-Chandra integral [McS21]. The proof of the Harish-Chandra formula provide by McSwiggen using the heat equation methods from Fegan’s 1978 paper [Feg78] seems very promising for deriving a connection to the character formula for the norm of the recoupling operator!
- Speyer [Spe05]
- Generalizations of Horn’s problem to the product of unitary matrices [AW97; Bel08; BK16]
- Bhatia [Bha01]
- History account offered by Daftuar and Hayden [DH05], and technical review article by Fulton [Ful00].
- Berline, Vergne, and Walter a pedagogical account of the triple role played by Horn’s inequalities in characterizing the existence of invariant vectors in tensor products, the eigenvalues of sums of Hermitian matrices, and intersections of Schubert varieties [BVW18]
- Fulton provides a comprehensive treatment of the connection between representation theory of $GL(d)$ and the geometry and (co)homology of the complex Grassmannians [Ful97]
- Prakash Belkale has contributed greatly to the theory surrounding Horn-type problems: for example, in [Bel04], Belkale provides a *geometric* proof of the equivalence between (i) the structure coefficients of the representation theory of $GL(d)$ (i.e. the LR-coefficients $c_{\mu\nu}^\lambda$), and (ii) the structure coefficients of the cohomology ring of the Grassmannian $Gr(d, n)$ of d -dimensional subspaces in $\mathbb{C}^d$. This proof is now complementary to the previously known algebraic proofs which rely on knowledge of the algebra of symmetric functions. This geometric approach taken by Belkale, which emphasizes the tangent spaces of the involved Schubert varieties, is later used by Belkale to provide a geometric proof of the saturation conjecture [Bel02].

B.2 Thompson’s conjecture

- Original (indirect) proof through the lense of random walks Klyachko [Kly00]
- The proof of Thompson’s conjecture in the style of Alekseev, Meinrenken, and Woodward is the “linearization paper” due to [AMW01].
- See also the related article due to Lamb which summarizes the ideas of [AMW01] while also providing more technical background [Lam05]
- Lu and Evens extend the proof technique of [AMW01] to a larger class of Lie groups [LE03]

- Alekseev, Podkopaeva, and Szenes [APS17a]
- Alekseev, Podkopaeva, and Szenes [APS17b]
- Alekseev and Meinrenken [AM07]
- Deforming the symplectic volume by the s parameter Alekseev et al. [Ale+18]
- **TODO:** track down pdfs for the original triple of papers by Thompson and Therianos titled “On the singular values of matrix products” I, II, and III from 1973
- **TODO:** track down pdfs for the original triple of papers by Thompson and Freede titled “On the eigenvalues of sums of Hermitian matrices” I, II, and III from 1973
- **TODO:** track down a pdf for Thompson’s 1988 “Matrix Spectral Inequalities” referenced by Klyachko
- Thompson and Therianos [TT73] present a number of interesting counter-examples to the question of whether the inequalities pertaining to the spectra of general matrices can be deduced from looking at the spectra of diagonal matrices (both in the additive and multiplicative cases).

B.3 $6j$ symbols

- For a comprehensive reference textbook for the $6j$ coefficients, see the textbook by Varshalovich, Moskalev, and Khersonskii [VMK88], specifically Chapter 9.
- A great summary by Roberts [Rob02]; also the original result for the asymptotics of the $6j$ symbols due to Roberts [Rob99]
- The masters essay/thesis of Backens (**TODO: check if the pdf Matthias sent to me is the final version**) [Bac10]
- The spectral quantum marginal problem and the $6j$ symbol for the symmetric group Christandl, Şahinoğlu, and Walter [CŞW18]
- Usage of Wigner’s $6j$ symbol to compute asymptotic error rates for the promise problem of distinguishing between two possible “change points” for a sequence of unknown pure states to transition from one unknown type to another [AH11]
- See also Coquereaux’s later paper which promotes the alternative, albeit equivalent, usage of the integer-valued gon and tet functions in place of the rational-valued theta (triangular) and irrational-valued $6j$ symbols [Coq23]. Most relevant for this project is [Coq23, Section IV D] which discusses the apparent lack of substantial generalization from $SU(2)$ to $SU(N)$ (“*apart from very few special cases (low dimension of irreps and no multiplicity) worked out by physicists at the time of the eightfold way, we are not aware of any published general result*”).
- The recent paper by Alekseev, Morozov, and Sleptsov (different Alekseev!) covers the case of some multiplicity free formulas for the quantum $6j$ symbols for $U_q(\mathfrak{sl}_N)$ [AMS20]
- Keppeler, Plätzer, and Sjodahl recently calculated $SU(d)$ $6j$ symbols with adjoint representations [KPS24] and a bit earlier [Alc+23] calculated $SU(d)$ $6j$ symbols with at least two fundamental representations [Alc+23].
- Representation theoretic proof of the Regge symmetries due to Boalch [Boa07]
- Asymptotics of quantum $6j$ symbols Taylor and Woodward [TW06]
- The integral formula for the square of the $SU(2)$ $6j$ symbol can be performed by an appropriate change of variables [FL03]. See also other papers by Laurent Freidel and collaborators that are relevant [FK00; FNR07; FKL10].

- Alcock-Zeilinger consider the case of $6j$ symbols for $SU(N)$ for wherein two opposing edges are labelled by fundamental irreps of $SU(N)$ so that all vertices are multiplicity one (or zero) [Alc18] TC: They also freely use the birdtrack notation which I enjoy.
- Only the tetrahedral symmetry of the $U_q(SL(N))$ $6j$ -symbol is known, the counterpart of the Regge symmetry is still unknown, although Morozov and Sleptsov conjectures the generalized Regge symmetry may be related to the eigenvalue conjecture, just as it is in the case when $N = 2$ [MS18].
- The graphical notion, specifically the use of an additional edge for multiplicity spaces can has been used for a long time, as early as [AB68; Mou84].
- The original Ponzano-Regge paper [PR68]
- Gurau derives the Ponzano-Regge asymptotic formula directly using Stirling’s approximation applied to Racah’s single-sum formula [Gur08]
- Bartlett and Nzaganya work with so-called *coherent loop states* or *Bohr-Sommerfeld states* to develop some semiclassical results concerning the inner products between such states (see also the talk on Youtube) [BN23]
- Aquilanti et al. provides an excellent introduction to subject of classical $6j$ symbols, their history, diagrammatics, presentations and asymptotics [Aqu+12]
- Marzuoli and Rasetti connects quantum computing models to the recoupling theory of $SU(2)$ angular momentum and also considers the semiclassical limit [MR05]
- Artamonov derives an explicit formula for the $d = 3$ case [Art23], and later to arbitrary d [Art24].
- There is a series of papers involving Alexey Sleptsov, starting with a computation of quantum $6j$ symbols for generic dimension d but with symmetric representations in terms of certain hypergeometric polynomials [MMS17], then with a discussion of the eigenvalue hypothesis and its potential connection to generalized Regge symmetries [MS18], then with a calculation of all multiplicity free quantum $6j$ symbols for arbitrary dimension [AMS20], and then finally with the discovery of novel “tug-the-hook” symmetries for quantum $6j$ symbols for all d [LS23].

B.4 Minkowski’s theorem and tetrahedron geometry

- Connecting group-valued moment maps to spherical, flat, and hyperbolic tetrahedron via Minkowski’s theorem from the reconstruction of flat tetrahedron Haggard, Han, and Riello. See also the comments in [HHR16, Section 11] on the utility of *quasi-poisson* (controlled failure of the Jacobi identity) manifolds and group-valued moment maps.
- Foundational results concerning the relationships between the edge lengths, face areas, dihedral and solid angles of a tetrahedron by Richardson [Ric02]
- Bianchi, Dona, and Speziale uses Minkowski’s theorem to parametrize convex polyhedra in $\mathbb{R}^3$ with specified face areas and the associated Kapovich-Millson phase space (i.e., the *space of shapes of polyhedra with fixed areas* and then considers its connection to loop quantum gravity via the usual Guillemin-Sternberg quantization commutes with reduction philosophy which associates invariants of the quantization of the unreduced phase space to the reduced phase space [BDS11] (notice the connection with [HHR16]).
- Regarding an algorithmic construction for the polyhedra appearing in Minkowski’s theorem and their complexity, see this post: [constructive-algorithm-for-minkowskis-theorem](#)
- Charles quantization of polygon spaces, tetrahedron and semiclassical limits [Cha10]

B.5 Quantization, Poisson geometry & quantum groups

- A popular textbook on quantum groups with an emphasis on the connection to Poisson manifolds, Lie bialgebras and their quantizations by Chari and Pressley [CP95]
- Course notes on Poisson geometry with an eye toward quantization by Ciccoli and Witkowski [CW06]
- The PhD thesis of Esposito (from the maths department here in Copenhagen) serves as a introduction to the world of Poisson momentum maps and the notion of a “quantum” momentum map as a deformation of the ordinary “classical” momentum map (where deformation is in the sense of deformation quantization) [Esp12].
- Alekseev [Ale97]
- Papers contained results pertaining to the generalization of Duistermaat-Heckman measures for Hamiltonian group actions with group-valued moment maps by Alekseev, Meinrenken, and Woodward [AMW99; AMW00]
- Kirillov and Reshetikhin [KR90]
- Octahedron Recurrence/Rule Robbins and Rumsey Jr [RR86] applied to the “Hive” ring Knutson, Tao, and Woodward [KTW03], and also Henriques and Kamnitzer [HK06]
- Henriques and Kamnitzer [HK05]
- A reference which contains the formula for the quantum dimension of a representation of $U_q(\mathfrak{sl}(d))$ labelled by λ , thanks Rinat! [Ais97]. See also [AM98] which clarifies the relation between various parameters. Apparently, the x, v, s parameters are apparently natural parameters to consider in the context of invariants of knots and links. The substitution mentioned in [AM98] is the following

$$s = e^{h/2}, \quad v = e^{-dh/2} = s^{-d}, \quad x = e^{-h/2d} = s^{-\frac{1}{d}}, \quad q = e^h = s^2.$$

This is consistent with [Res87] in the sense that

$$[k] := \frac{s^{+k} - s^{-k}}{s^{+1} - s^{-1}} = \frac{q^{+k/2} - q^{-k/2}}{q^{+1/2} - q^{-1/2}}. \quad (155)$$

- Reshetikhin develops the diagrammatics of the category of representations of $U_q(\mathfrak{g})$ motivated by connections to the developing theory of polynomial invariants of knots and links [Res87].
- Moak investigates a q -Stirling’s formula for the logarithm of the q -factorial [Moa84].

B.6 Eigenvalue inequalities

- The general subject of multivariate trace inequalities involving exponentials of sums of Hermitian and products of their exponentials is, of course, intimately related. See for instance the paper by Sutter, Berta, and Tomamichel [SBT17].
- Also the article [FT14] by Forrester and Thompson which describes the connection between Thompson’s inequality and hyperbolic geometry quite explicitly.
- Friedland and Gaubert considers submodular functions associated to operator monotones evaluated on different principal submatrices of a single Hermitian matrix [FG13]. While it is not clear to be yet whether or not one can relate partial sums of matrices to principal submatrices, [FG13] references a paper by Bouhtou, Gaubert, and Sagnol [BGS08] which proves the following family of inequalities for all positive semidefinite $A, B, C \geq 0$ (and $p \in [0, 1]$) [BGS08, Lemma 5.3]:

$$\sum_j \lambda_j (A + B)^p + \sum_j \lambda_j (B + C)^p \geq \sum_j \lambda_j (B)^p + \sum_j \lambda_j (A + B + C)^p. \quad (156)$$

- Audenaert, Hiai, and Petz consider a similar form of “strongly subadditive” functions [AHP10]
- For multivariate inequalities based on idempotents of the symmetric group algebra see Huber’s paper [Hub20].
- For spectral inequalities derived using semidefinite programming, see Huber and Wyderka’s paper [HW22].
- Paksoy, Turkmen, and Zhang produce a large family of polynomial inequalities for positive semidefinite $A, B, C \geq 0$ using the notion of a “generalized matrix function” [PTZ14] (see also [HM21; MM65])

B.7 Uncategorized

- Elementary introduction to toy models of three-dimensional quantum space [Bar03]
- Regge and Williams [RW00] a relatively recent review of discrete approaches to gravity. The essential idea, in contrast to the differential geometry of general relativity, is to model curvature discretely instead of continuously. In particular, one can think of a triangulation of the surface of a sphere; the curvature of the sphere is encoded in the triangulation when a configuration of three (or more) triangles meet at a common vertex with sum of interior angles deviating from π . It is in this sense that Regge and Williams refers to these types of models as “piecewise-flat”. Contained is an overview of various attempts to generalize discrete theories of three-dimensional gravity to four dimensions [RW00, Section III].